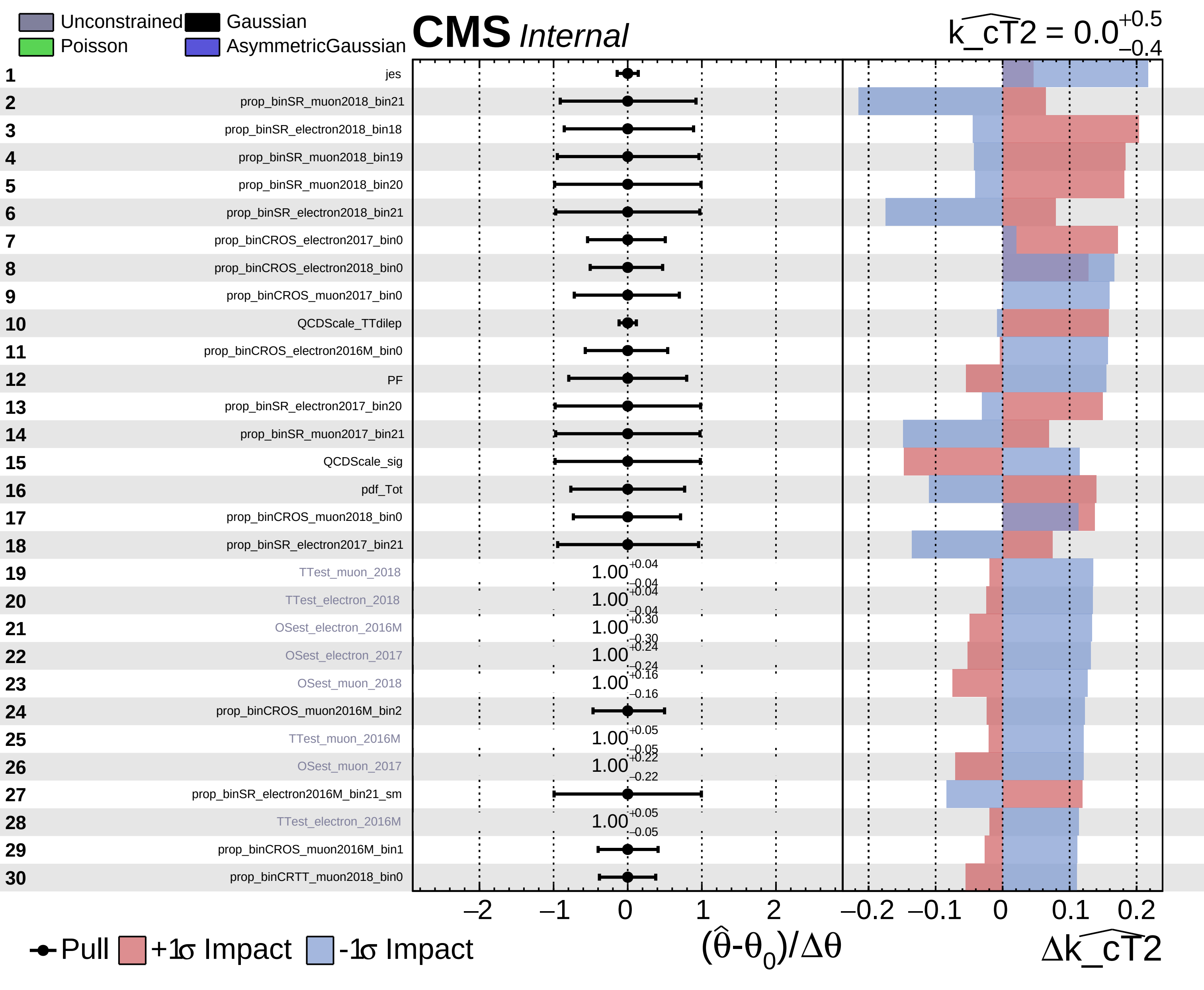
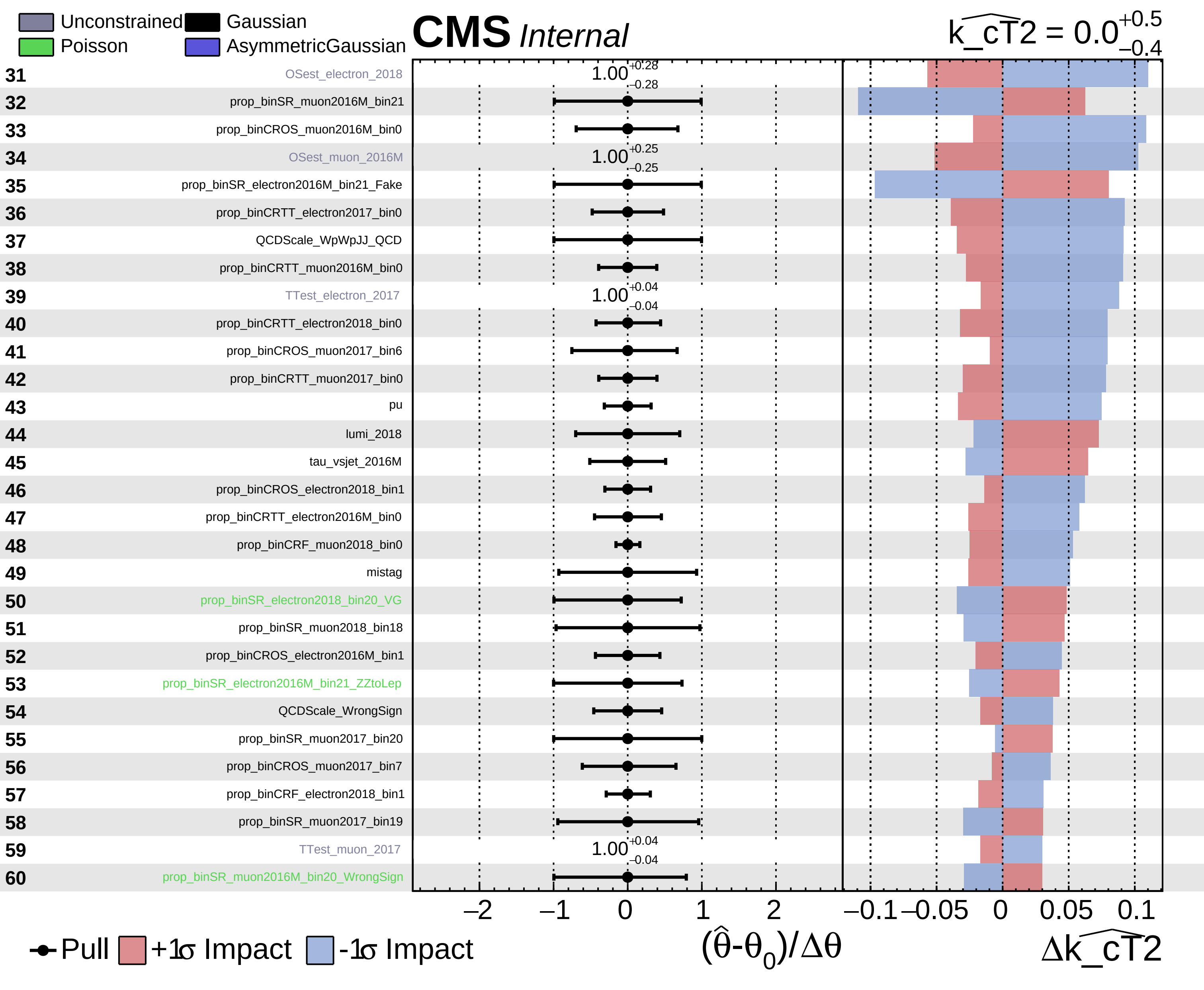
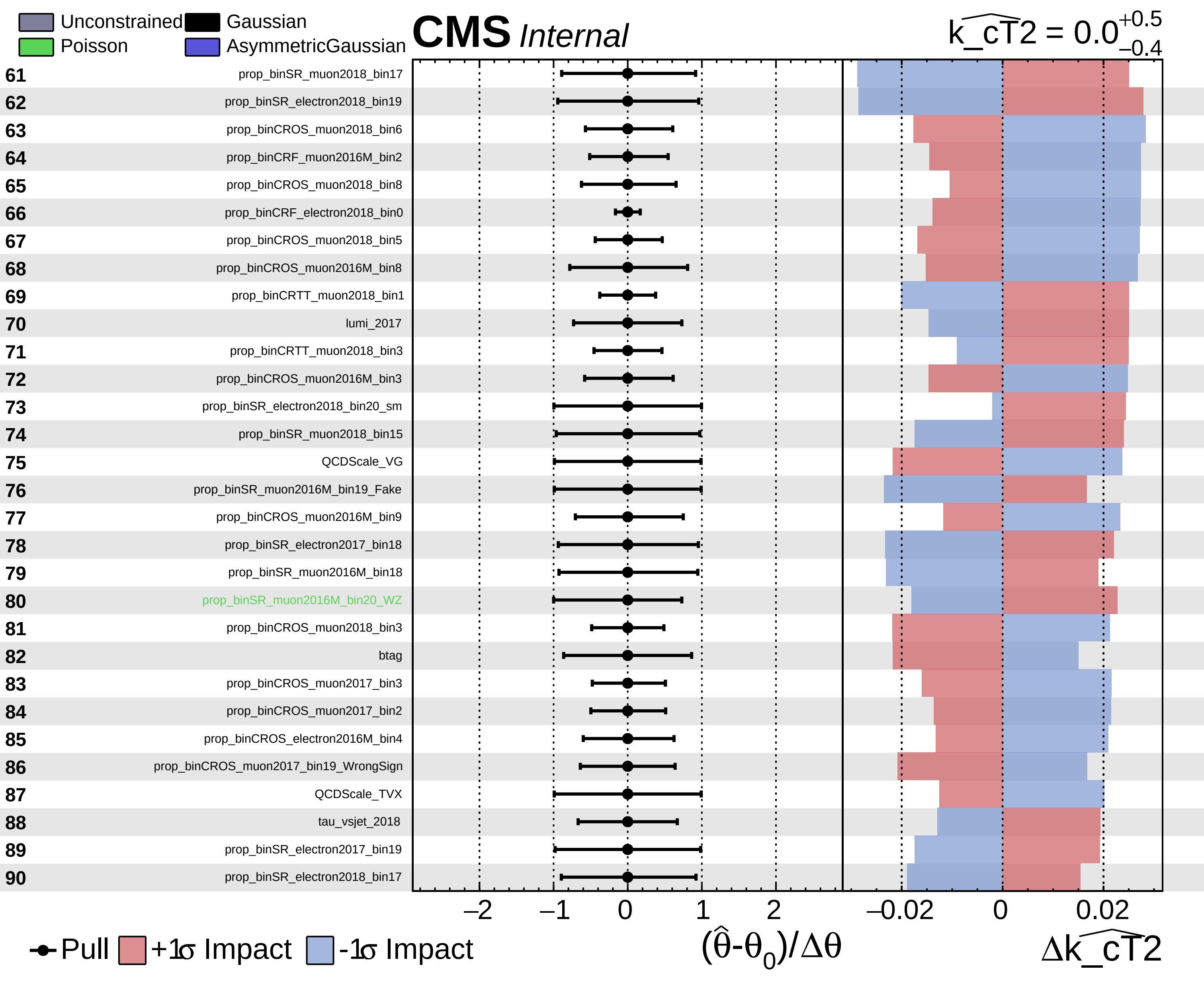
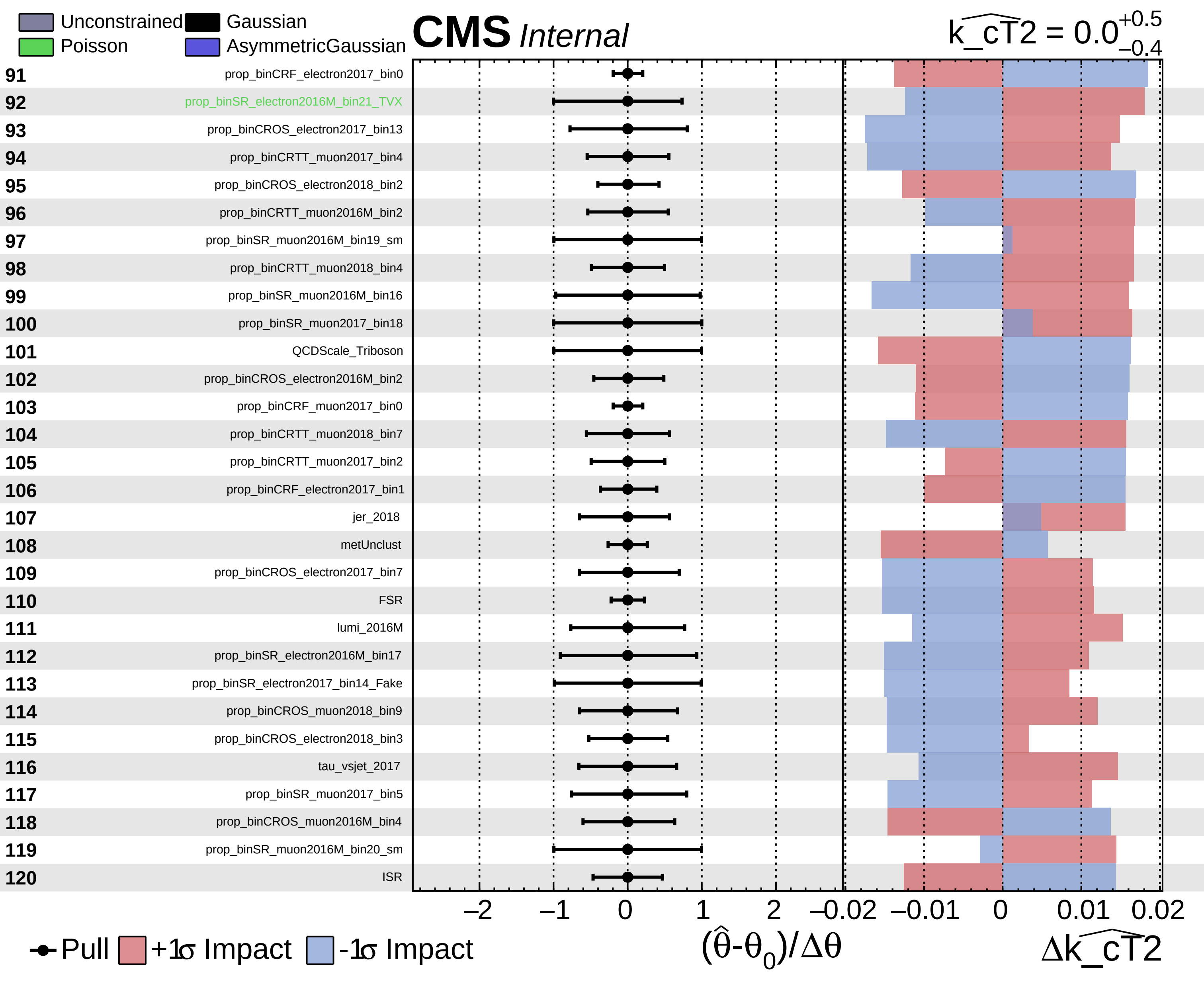
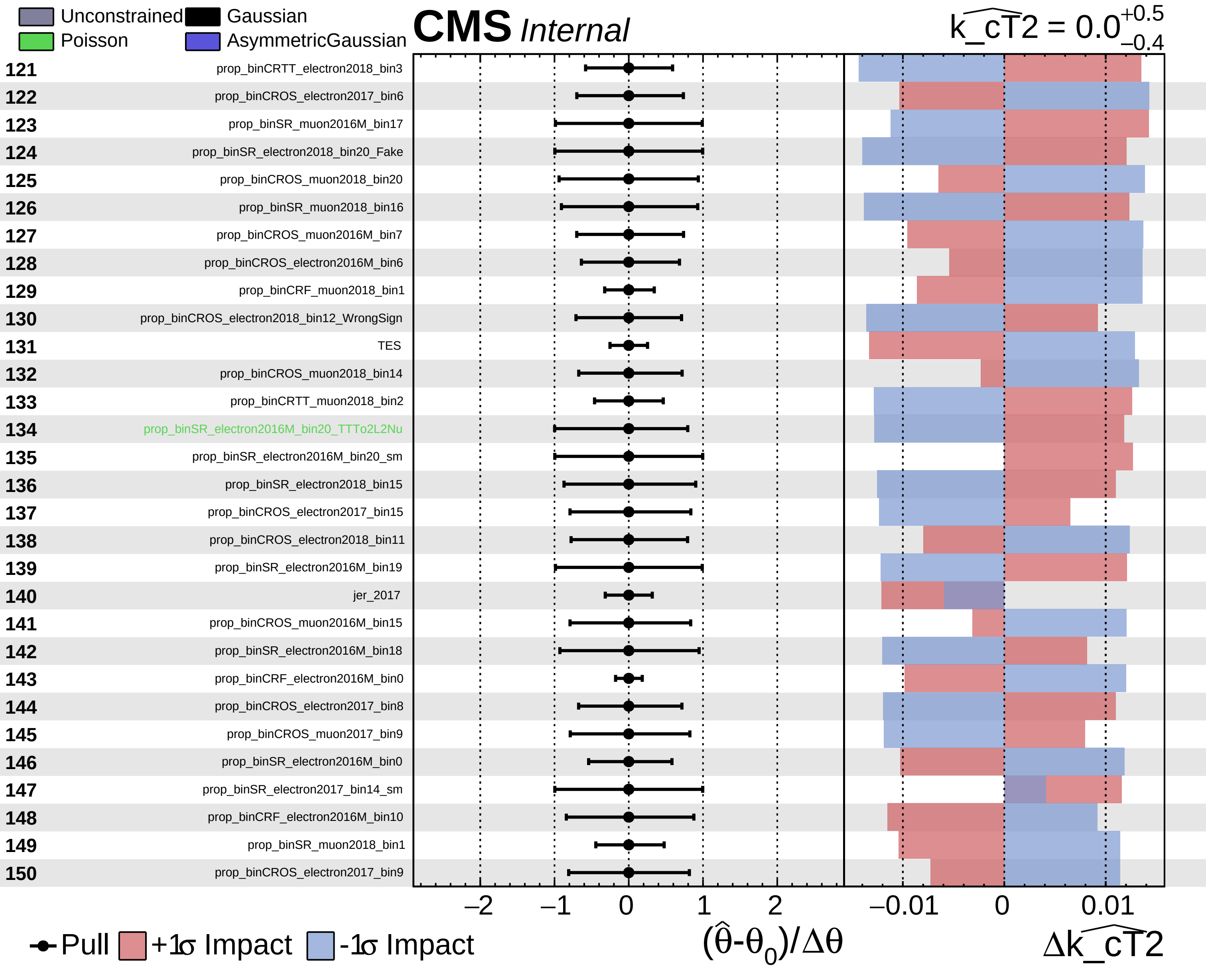

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
1 jes
2 prop_binSR_muon2018_bin21
3 prop_binSR_electron2018_bin18
4 prop_binSR_muon2018_bin19
5 prop_binSR_muon2018_bin20
6 prop_binSR_electron2018_bin21
7 prop_binCROS_electron2017_bin0
8 prop_binCROS_electron2018_bin0
9 prop_binCROS_muon2017_bin0
10 QCDScale_TTdilep
11 prop_binCROS_electron2016M_bin0
12 PF
13 prop_binSR_electron2017_bin20
14 prop_binSR_muon2017_bin21
15 QCDScale_sig
16 pdf_Tot
17 prop_binCROS_muon2018_bin0
18 prop_binSR_electron2017_bin21
19 TTest_muon_2018 1.00 +0.04 −0.04
20 TTest_electron_2018 1.00 +0.04 −0.04
21 OSest_electron_2016M 1.00 +0.30 −0.30
22 OSest_electron_2017 1.00 +0.24 −0.24
23 OSest_muon_2018 1.00 +0.16 −0.16
24 prop_binCROS_muon2016M_bin2
25 TTest_muon_2016M 1.00 +0.05 −0.05
26 OSest_muon_2017 1.00 +0.22 −0.22
27 prop_binSR_electron2016M_bin21_sm
28 TTest_electron_2016M 1.00 +0.05 −0.05
29 prop_binCROS_muon2016M_bin1
30 prop_binCRTT_muon2018_bin0
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



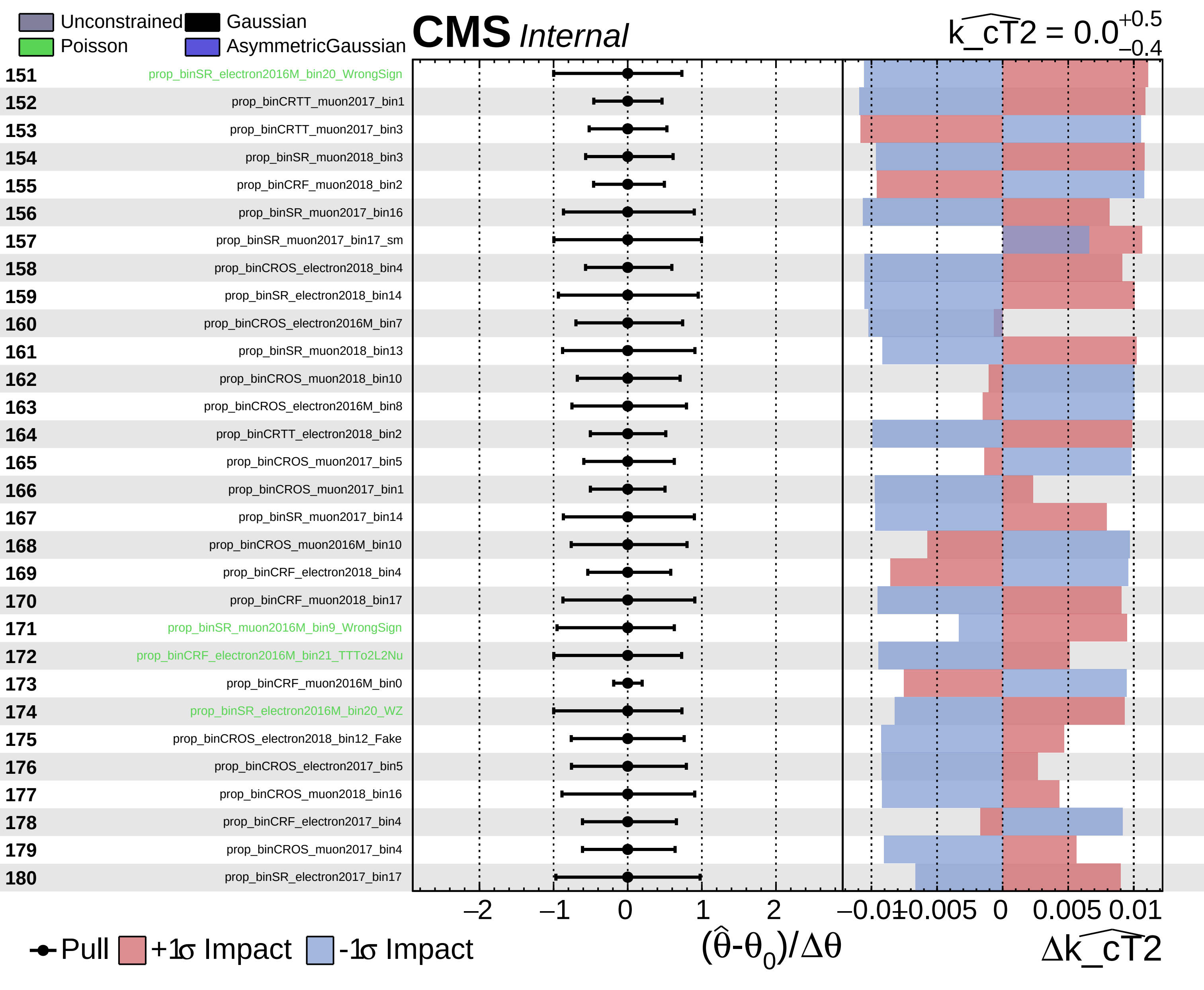
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
31 OSest_electron_2018 1.00 +0.28 −0.28
32 prop_binSR_muon2016M_bin21
33 prop_binCROS_muon2016M_bin0
34 OSest_muon_2016M 1.00 +0.25 −0.25
35 prop_binSR_electron2016M_bin21_Fake
36 prop_binCRTT_electron2017_bin0
37 QCDScale_WpWpJJ_QCD
38 prop_binCRTT_muon2016M_bin0
39 TTest_electron_2017 1.00 +0.04 −0.04
40 prop_binCRTT_electron2018_bin0
41 prop_binCROS_muon2017_bin6
42 prop_binCRTT_muon2017_bin0
43 pu
44 lumi_2018
45 tau_vsjet_2016M
46 prop_binCROS_electron2018_bin1
47 prop_binCRTT_electron2016M_bin0
48 prop_binCRF_muon2018_bin0
49 mistag
50 prop_binSR_electron2018_bin20_VG
51 prop_binSR_muon2018_bin18
52 prop_binCROS_electron2016M_bin1
53 prop_binSR_electron2016M_bin21_ZZtoLep
54 QCDScale_WrongSign
55 prop_binSR_muon2017_bin20
56 prop_binCROS_muon2017_bin7
57 prop_binCRF_electron2018_bin1
58 prop_binSR_muon2017_bin19
59 TTest_muon_2017 1.00 +0.04 −0.04
60 prop_binSR_muon2016M_bin20_WrongSign
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
61 prop_binSR_muon2018_bin17
62 prop_binSR_electron2018_bin19
63 prop_binCROS_muon2018_bin6
64 prop_binCRF_muon2016M_bin2
65 prop_binCROS_muon2018_bin8
66 prop_binCRF_electron2018_bin0
67 prop_binCROS_muon2018_bin5
68 prop_binCROS_muon2016M_bin8
69 prop_binCRTT_muon2018_bin1
70 lumi_2017
71 prop_binCRTT_muon2018_bin3
72 prop_binCROS_muon2016M_bin3
73 prop_binSR_electron2018_bin20_sm
74 prop_binSR_muon2018_bin15
75 QCDScale_VG
76 prop_binSR_muon2016M_bin19_Fake
77 prop_binCROS_muon2016M_bin9
78 prop_binSR_electron2017_bin18
79 prop_binSR_muon2016M_bin18
80 prop_binSR_muon2016M_bin20_WZ
81 prop_binCROS_muon2018_bin3
82 btag
83 prop_binCROS_muon2017_bin3
84 prop_binCROS_muon2017_bin2
85 prop_binCROS_electron2016M_bin4
86 prop_binCROS_muon2017_bin19_WrongSign
87 QCDScale_TVX
88 tau_vsjet_2018
89 prop_binSR_electron2017_bin19
90 prop_binSR_electron2018_bin17
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

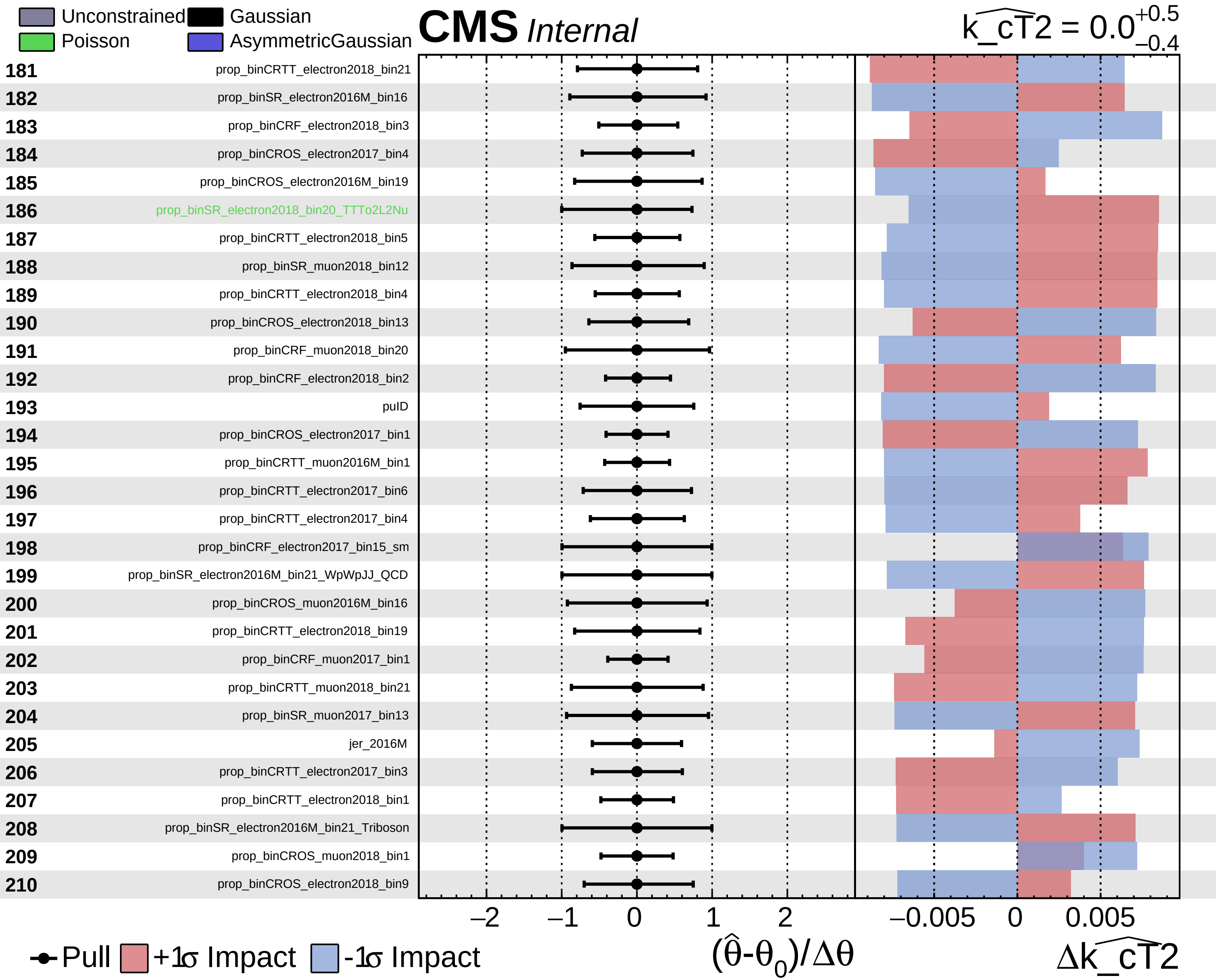

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
91 prop_binCRF_electron2017_bin0
92 prop_binSR_electron2016M_bin21_TVX
93 prop_binCROS_electron2017_bin13
94 prop_binCRTT_muon2017_bin4
95 prop_binCROS_electron2018_bin2
96 prop_binCRTT_muon2016M_bin2
97 prop_binSR_muon2016M_bin19_sm
98 prop_binCRTT_muon2018_bin4
99 prop_binSR_muon2016M_bin16
100 prop_binSR_muon2017_bin18
101 QCDScale_Triboson
102 prop_binCROS_electron2016M_bin2
103 prop_binCRF_muon2017_bin0
104 prop_binCRTT_muon2018_bin7
105 prop_binCRTT_muon2017_bin2
106 prop_binCRF_electron2017_bin1
107 jer_2018
108 metUnclust
109 prop_binCROS_electron2017_bin7
110 FSR
111 lumi_2016M
112 prop_binSR_electron2016M_bin17
113 prop_binSR_electron2017_bin14_Fake
114 prop_binCROS_muon2018_bin9
115 prop_binCROS_electron2018_bin3
116 tau_vsjet_2017
117 prop_binSR_muon2017_bin5
118 prop_binCROS_muon2016M_bin4
119 prop_binSR_muon2016M_bin20_sm
120 ISR
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



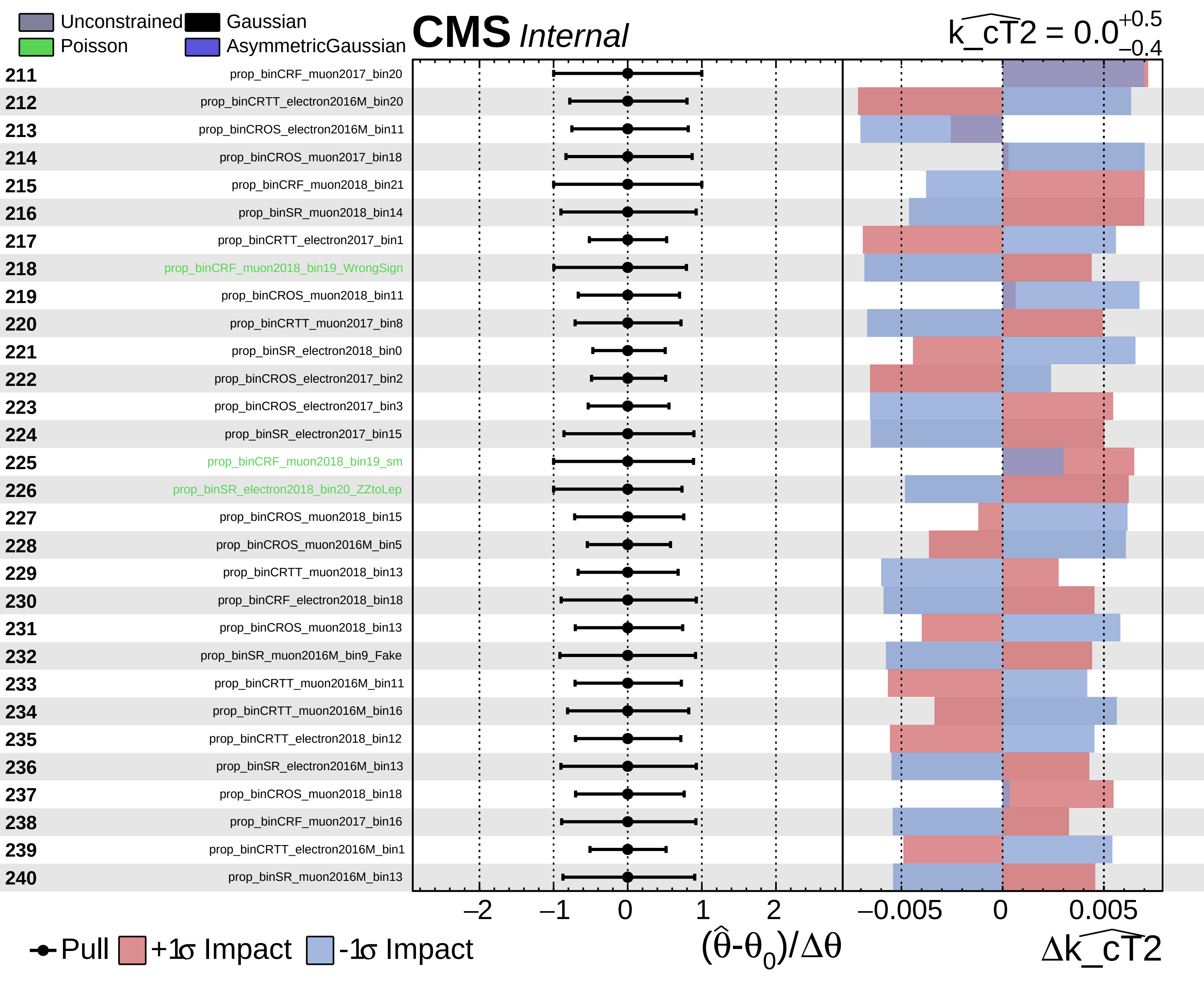
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
121 prop_binCRTT_electron2018_bin3
122 prop_binCROS_electron2017_bin6
123 prop_binSR_muon2016M_bin17
124 prop_binSR_electron2018_bin20_Fake
125 prop_binCROS_muon2018_bin20
126 prop_binSR_muon2018_bin16
127 prop_binCROS_muon2016M_bin7
128 prop_binCROS_electron2016M_bin6
129 prop_binCRF_muon2018_bin1
130 prop_binCROS_electron2018_bin12_WrongSign
131 TES
132 prop_binCROS_muon2018_bin14
133 prop_binCRTT_muon2018_bin2
134 prop_binSR_electron2016M_bin20_TTTo2L2Nu
135 prop_binSR_electron2016M_bin20_sm
136 prop_binSR_electron2018_bin15
137 prop_binCROS_electron2017_bin15
138 prop_binCROS_electron2018_bin11
139 prop_binSR_electron2016M_bin19
140 jer_2017
141 prop_binCROS_muon2016M_bin15
142 prop_binSR_electron2016M_bin18
143 prop_binCRF_electron2016M_bin0
144 prop_binCROS_electron2017_bin8
145 prop_binCROS_muon2017_bin9
146 prop_binSR_electron2016M_bin0
147 prop_binSR_electron2017_bin14_sm
148 prop_binCRF_electron2016M_bin10
149 prop_binSR_muon2018_bin1
150 prop_binCROS_electron2017_bin9
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



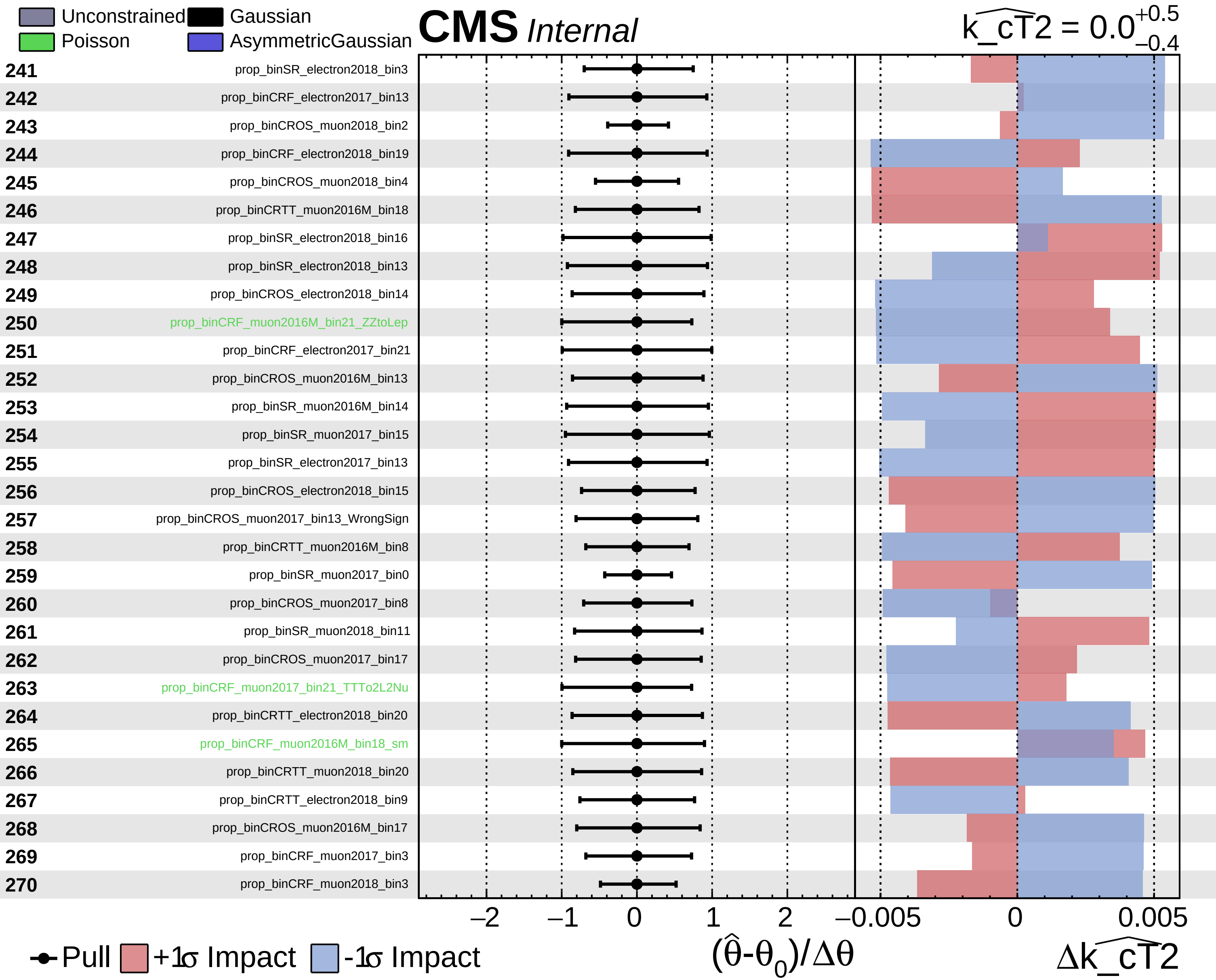
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
151 prop_binSR_electron2016M_bin20_WrongSign
152 prop_binCRTT_muon2017_bin1
153 prop_binCRTT_muon2017_bin3
154 prop_binSR_muon2018_bin3
155 prop_binCRF_muon2018_bin2
156 prop_binSR_muon2017_bin16
157 prop_binSR_muon2017_bin17_sm
158 prop_binCROS_electron2018_bin4
159 prop_binSR_electron2018_bin14
160 prop_binCROS_electron2016M_bin7
161 prop_binSR_muon2018_bin13
162 prop_binCROS_muon2018_bin10
163 prop_binCROS_electron2016M_bin8
164 prop_binCRTT_electron2018_bin2
165 prop_binCROS_muon2017_bin5
166 prop_binCROS_muon2017_bin1
167 prop_binSR_muon2017_bin14
168 prop_binCROS_muon2016M_bin10
169 prop_binCRF_electron2018_bin4
170 prop_binCRF_muon2018_bin17
171 prop_binSR_muon2016M_bin9_WrongSign
172 prop_binCRF_electron2016M_bin21_TTTo2L2Nu
173 prop_binCRF_muon2016M_bin0
174 prop_binSR_electron2016M_bin20_WZ
175 prop_binCROS_electron2018_bin12_Fake
176 prop_binCROS_electron2017_bin5
177 prop_binCROS_muon2018_bin16
178 prop_binCRF_electron2017_bin4
179 prop_binCROS_muon2017_bin4
180 prop_binSR_electron2017_bin17
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

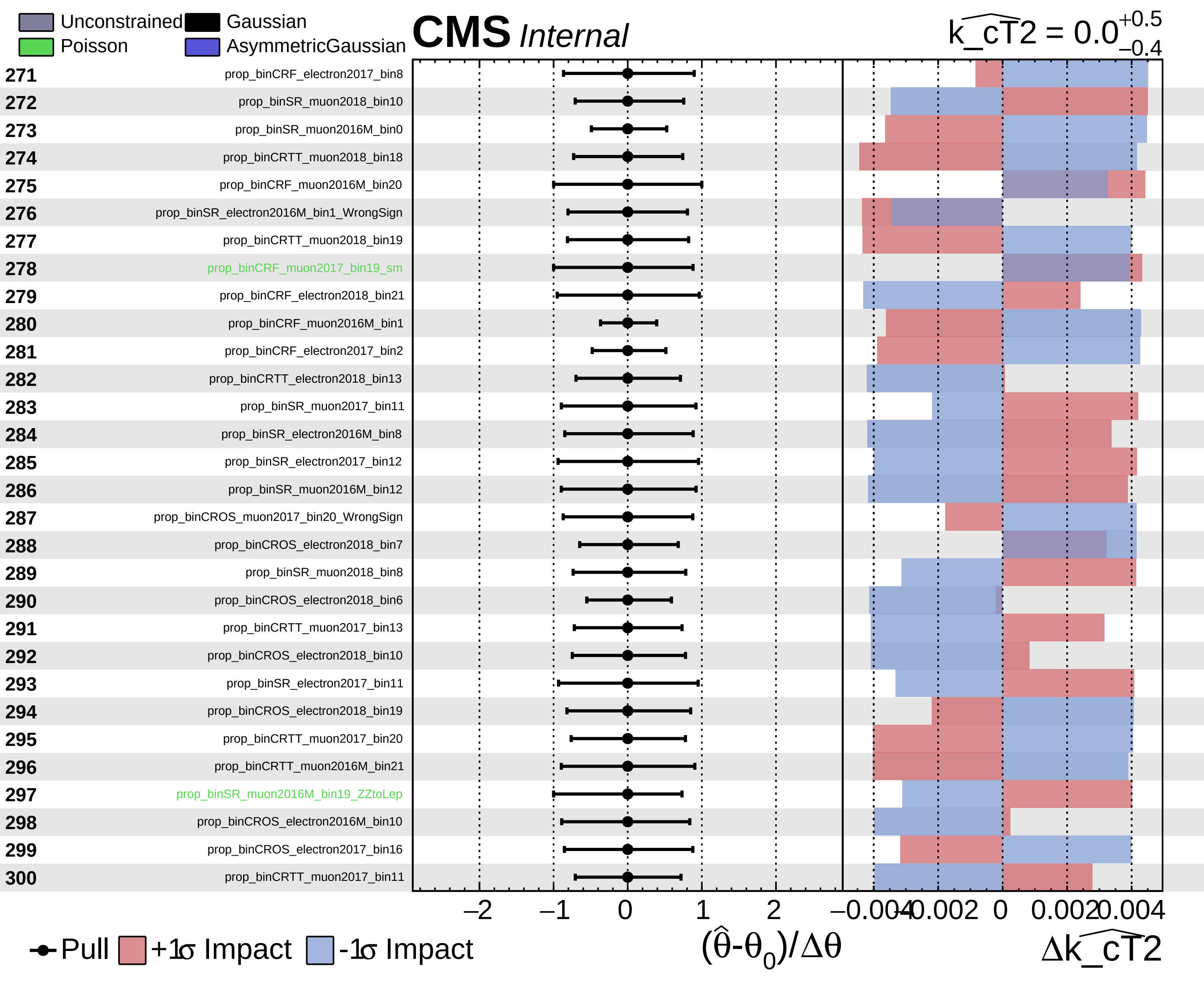


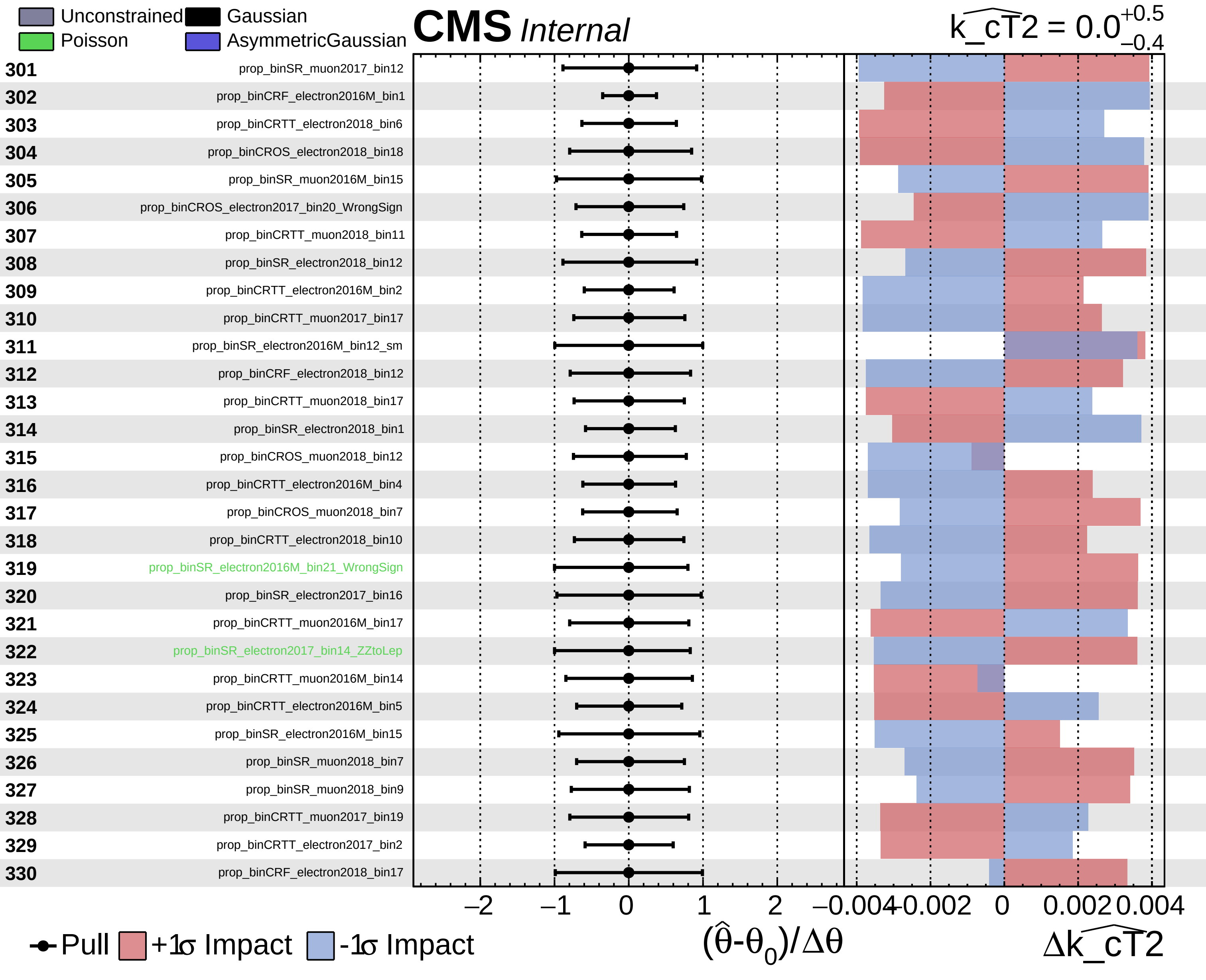
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
181 prop_binCRTT_electron2018_bin21
182 prop_binSR_electron2016M_bin16
183 prop_binCRF_electron2018_bin3
184 prop_binCROS_electron2017_bin4
185 prop_binCROS_electron2016M_bin19
186 prop_binSR_electron2018_bin20_TTTo2L2Nu
187 prop_binCRTT_electron2018_bin5
188 prop_binSR_muon2018_bin12
189 prop_binCRTT_electron2018_bin4
190 prop_binCROS_electron2018_bin13
191 prop_binCRF_muon2018_bin20
192 prop_binCRF_electron2018_bin2
193 puID
194 prop_binCROS_electron2017_bin1
195 prop_binCRTT_muon2016M_bin1
196 prop_binCRTT_electron2017_bin6
197 prop_binCRTT_electron2017_bin4
198 prop_binCRF_electron2017_bin15_sm
199 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
200 prop_binCROS_muon2016M_bin16
201 prop_binCRTT_electron2018_bin19
202 prop_binCRF_muon2017_bin1
203 prop_binCRTT_muon2018_bin21
204 prop_binSR_muon2017_bin13
205 jer_2016M
206 prop_binCRTT_electron2017_bin3
207 prop_binCRTT_electron2018_bin1
208 prop_binSR_electron2016M_bin21_Triboson
209 prop_binCROS_muon2018_bin1
210 prop_binCROS_electron2018_bin9
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



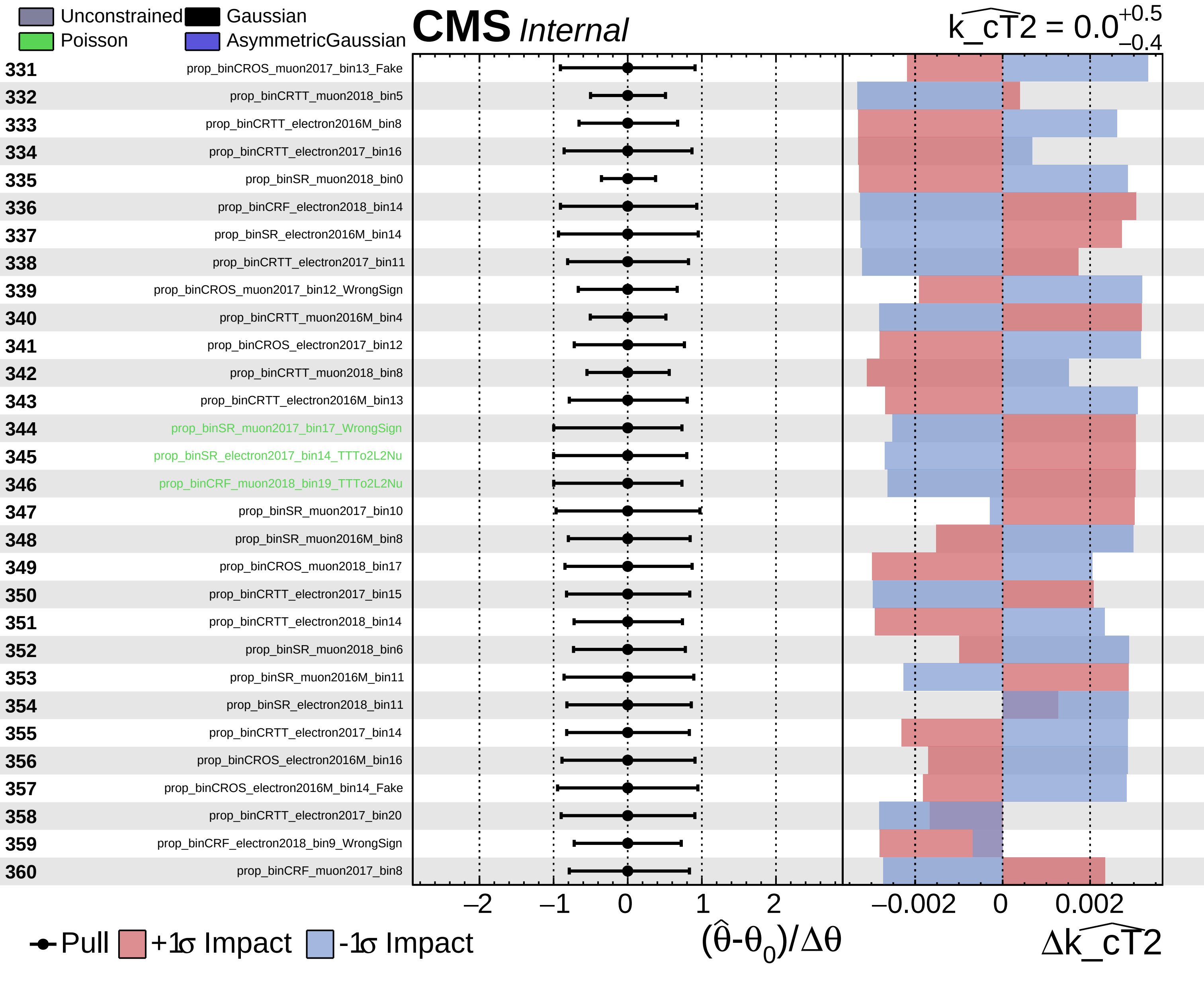
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
211 prop_binCRF_muon2017_bin20
212 prop_binCRTT_electron2016M_bin20
213 prop_binCROS_electron2016M_bin11
214 prop_binCROS_muon2017_bin18
215 prop_binCRF_muon2018_bin21
216 prop_binSR_muon2018_bin14
217 prop_binCRTT_electron2017_bin1
218 prop_binCRF_muon2018_bin19_WrongSign
219 prop_binCROS_muon2018_bin11
220 prop_binCRTT_muon2017_bin8
221 prop_binSR_electron2018_bin0
222 prop_binCROS_electron2017_bin2
223 prop_binCROS_electron2017_bin3
224 prop_binSR_electron2017_bin15
225 prop_binCRF_muon2018_bin19_sm
226 prop_binSR_electron2018_bin20_ZZtoLep
227 prop_binCROS_muon2018_bin15
228 prop_binCROS_muon2016M_bin5
229 prop_binCRTT_muon2018_bin13
230 prop_binCRF_electron2018_bin18
231 prop_binCROS_muon2018_bin13
232 prop_binSR_muon2016M_bin9_Fake
233 prop_binCRTT_muon2016M_bin11
234 prop_binCRTT_muon2016M_bin16
235 prop_binCRTT_electron2018_bin12
236 prop_binSR_electron2016M_bin13
237 prop_binCROS_muon2018_bin18
238 prop_binCRF_muon2017_bin16
239 prop_binCRTT_electron2016M_bin1
240 prop_binSR_muon2016M_bin13
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
241 prop_binSR_electron2018_bin3
242 prop_binCRF_electron2017_bin13
243 prop_binCROS_muon2018_bin2
244 prop_binCRF_electron2018_bin19
245 prop_binCROS_muon2018_bin4
246 prop_binCRTT_muon2016M_bin18
247 prop_binSR_electron2018_bin16
248 prop_binSR_electron2018_bin13
249 prop_binCROS_electron2018_bin14
250 prop_binCRF_muon2016M_bin21_ZZtoLep
251 prop_binCRF_electron2017_bin21
252 prop_binCROS_muon2016M_bin13
253 prop_binSR_muon2016M_bin14
254 prop_binSR_muon2017_bin15
255 prop_binSR_electron2017_bin13
256 prop_binCROS_electron2018_bin15
257 prop_binCROS_muon2017_bin13_WrongSign
258 prop_binCRTT_muon2016M_bin8
259 prop_binSR_muon2017_bin0
260 prop_binCROS_muon2017_bin8
261 prop_binSR_muon2018_bin11
262 prop_binCROS_muon2017_bin17
263 prop_binCRF_muon2017_bin21_TTTo2L2Nu
264 prop_binCRTT_electron2018_bin20
265 prop_binCRF_muon2016M_bin18_sm
266 prop_binCRTT_muon2018_bin20
267 prop_binCRTT_electron2018_bin9
268 prop_binCROS_muon2016M_bin17
269 prop_binCRF_muon2017_bin3
270 prop_binCRF_muon2018_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

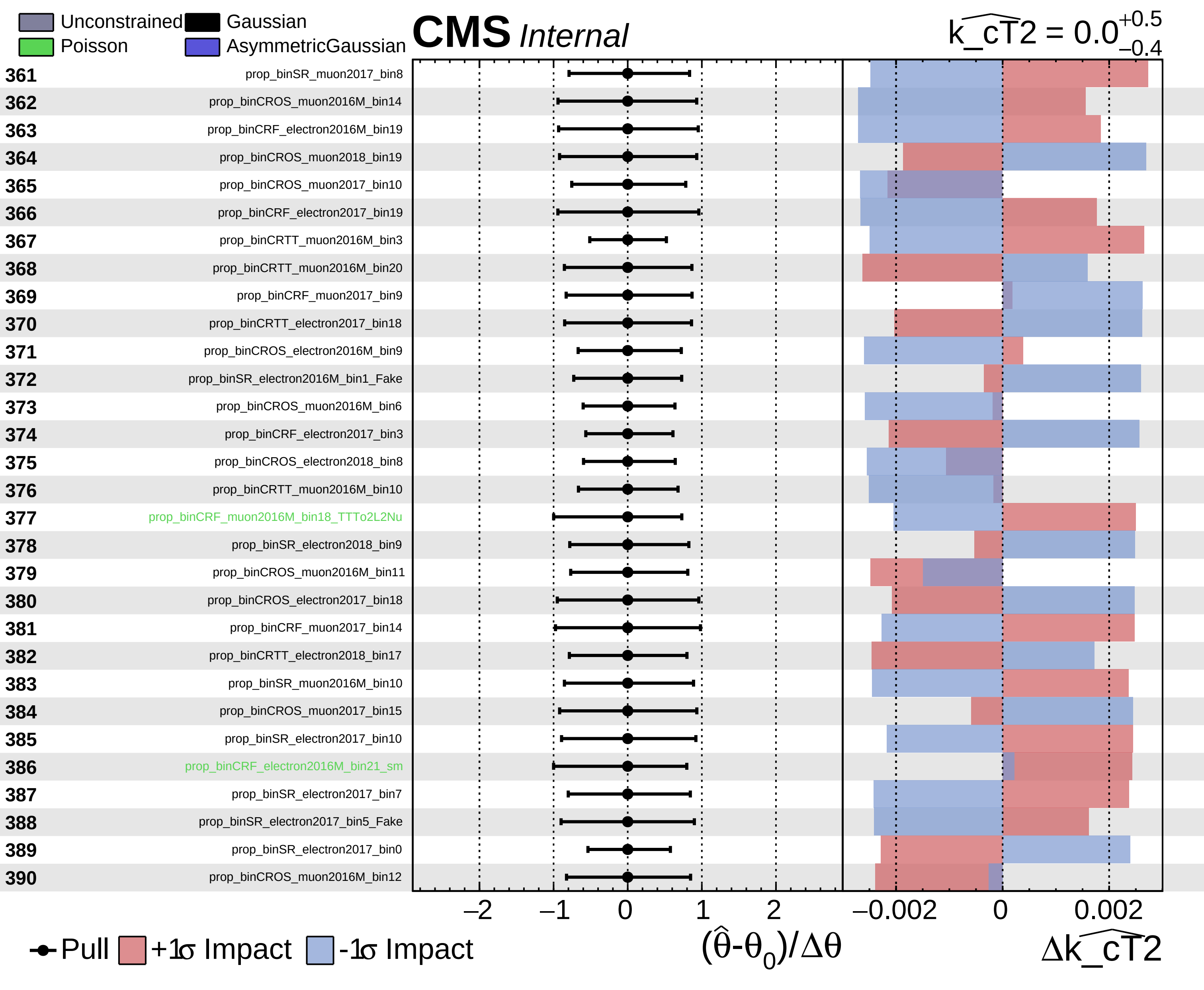

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
271 prop_binCRF_electron2017_bin8
272 prop_binSR_muon2018_bin10
273 prop_binSR_muon2016M_bin0
274 prop_binCRTT_muon2018_bin18
275 prop_binCRF_muon2016M_bin20
276 prop_binSR_electron2016M_bin1_WrongSign
277 prop_binCRTT_muon2018_bin19
278 prop_binCRF_muon2017_bin19_sm
279 prop_binCRF_electron2018_bin21
280 prop_binCRF_muon2016M_bin1
281 prop_binCRF_electron2017_bin2
282 prop_binCRTT_electron2018_bin13
283 prop_binSR_muon2017_bin11
284 prop_binSR_electron2016M_bin8
285 prop_binSR_electron2017_bin12
286 prop_binSR_muon2016M_bin12
287 prop_binCROS_muon2017_bin20_WrongSign
288 prop_binCROS_electron2018_bin7
289 prop_binSR_muon2018_bin8
290 prop_binCROS_electron2018_bin6
291 prop_binCRTT_muon2017_bin13
292 prop_binCROS_electron2018_bin10
293 prop_binSR_electron2017_bin11
294 prop_binCROS_electron2018_bin19
295 prop_binCRTT_muon2017_bin20
296 prop_binCRTT_muon2016M_bin21
297 prop_binSR_muon2016M_bin19_ZZtoLep
298 prop_binCROS_electron2016M_bin10
299 prop_binCROS_electron2017_bin16
300 prop_binCRTT_muon2017_bin11
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

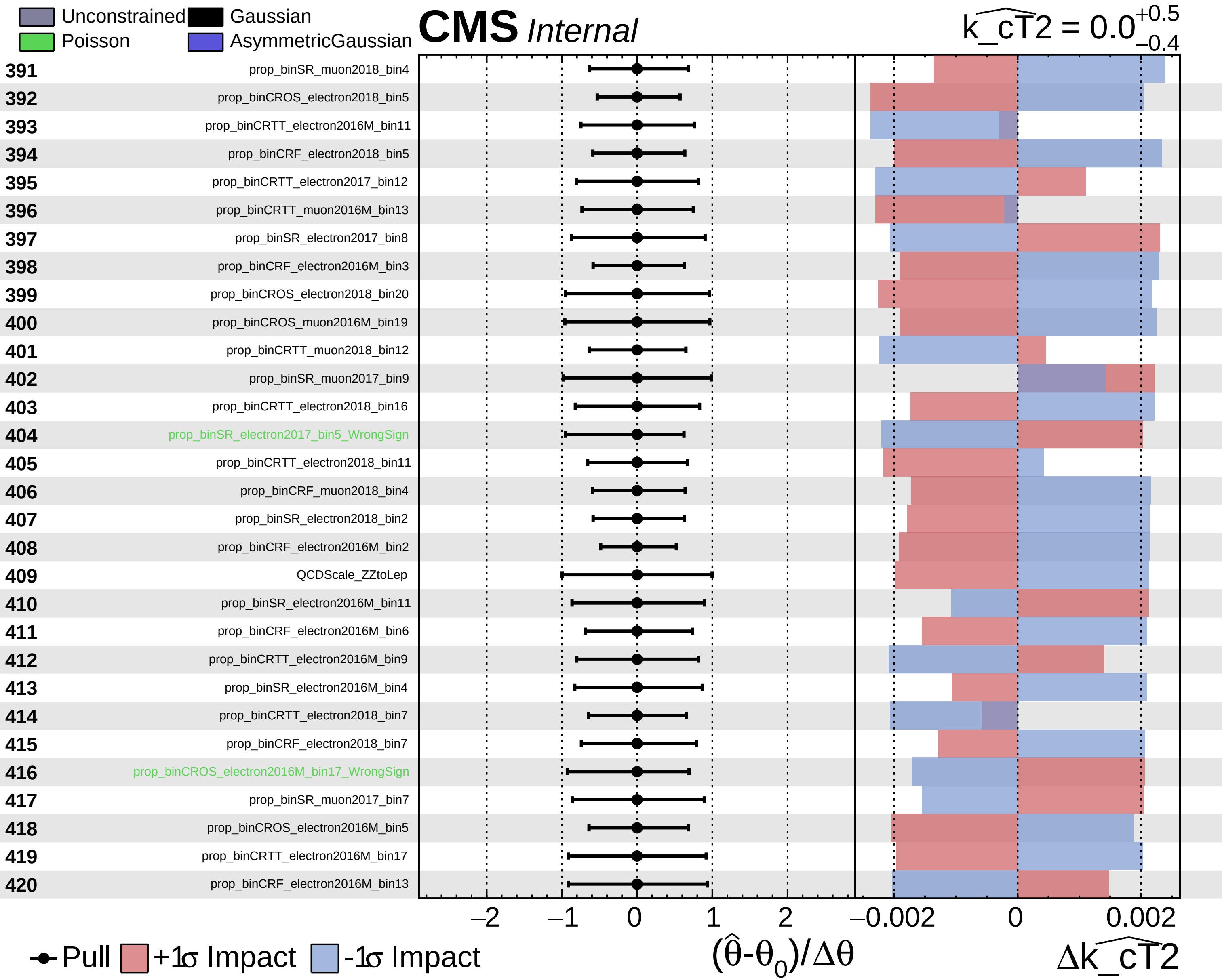


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
301 prop_binSR_muon2017_bin12
302 prop_binCRF_electron2016M_bin1
303 prop_binCRTT_electron2018_bin6
304 prop_binCROS_electron2018_bin18
305 prop_binSR_muon2016M_bin15
306 prop_binCROS_electron2017_bin20_WrongSign
307 prop_binCRTT_muon2018_bin11
308 prop_binSR_electron2018_bin12
309 prop_binCRTT_electron2016M_bin2
310 prop_binCRTT_muon2017_bin17
311 prop_binSR_electron2016M_bin12_sm
312 prop_binCRF_electron2018_bin12
313 prop_binCRTT_muon2018_bin17
314 prop_binSR_electron2018_bin1
315 prop_binCROS_muon2018_bin12
316 prop_binCRTT_electron2016M_bin4
317 prop_binCROS_muon2018_bin7
318 prop_binCRTT_electron2018_bin10
319 prop_binSR_electron2016M_bin21_WrongSign
320 prop_binSR_electron2017_bin16
321 prop_binCRTT_muon2016M_bin17
322 prop_binSR_electron2017_bin14_ZZtoLep
323 prop_binCRTT_muon2016M_bin14
324 prop_binCRTT_electron2016M_bin5
325 prop_binSR_electron2016M_bin15
326 prop_binSR_muon2018_bin7
327 prop_binSR_muon2018_bin9
328 prop_binCRTT_muon2017_bin19
329 prop_binCRTT_electron2017_bin2
330 prop_binCRF_electron2018_bin17
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

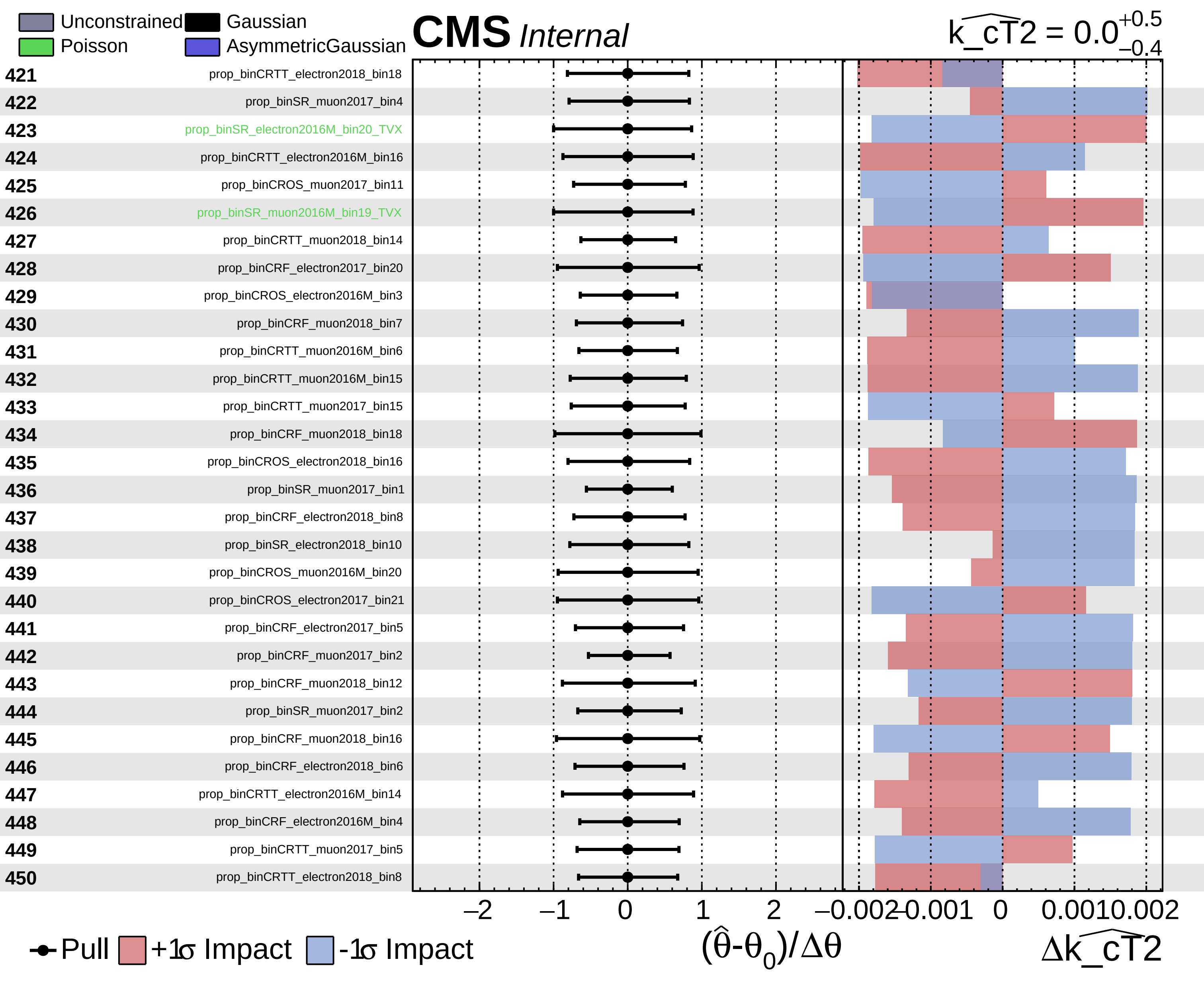


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
331 prop_binCROS_muon2017_bin13_Fake
332 prop_binCRTT_muon2018_bin5
333 prop_binCRTT_electron2016M_bin8
334 prop_binCRTT_electron2017_bin16
335 prop_binSR_muon2018_bin0
336 prop_binCRF_electron2018_bin14
337 prop_binSR_electron2016M_bin14
338 prop_binCRTT_electron2017_bin11
339 prop_binCROS_muon2017_bin12_WrongSign
340 prop_binCRTT_muon2016M_bin4
341 prop_binCROS_electron2017_bin12
342 prop_binCRTT_muon2018_bin8
343 prop_binCRTT_electron2016M_bin13
344 prop_binSR_muon2017_bin17_WrongSign
345 prop_binSR_electron2017_bin14_TTTo2L2Nu
346 prop_binCRF_muon2018_bin19_TTTo2L2Nu
347 prop_binSR_muon2017_bin10
348 prop_binSR_muon2016M_bin8
349 prop_binCROS_muon2018_bin17
350 prop_binCRTT_electron2017_bin15
351 prop_binCRTT_electron2018_bin14
352 prop_binSR_muon2018_bin6
353 prop_binSR_muon2016M_bin11
354 prop_binSR_electron2018_bin11
355 prop_binCRTT_electron2017_bin14
356 prop_binCROS_electron2016M_bin16
357 prop_binCROS_electron2016M_bin14_Fake
358 prop_binCRTT_electron2017_bin20
359 prop_binCRF_electron2018_bin9_WrongSign
360 prop_binCRF_muon2017_bin8
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta \widehat{k\_cT2}$

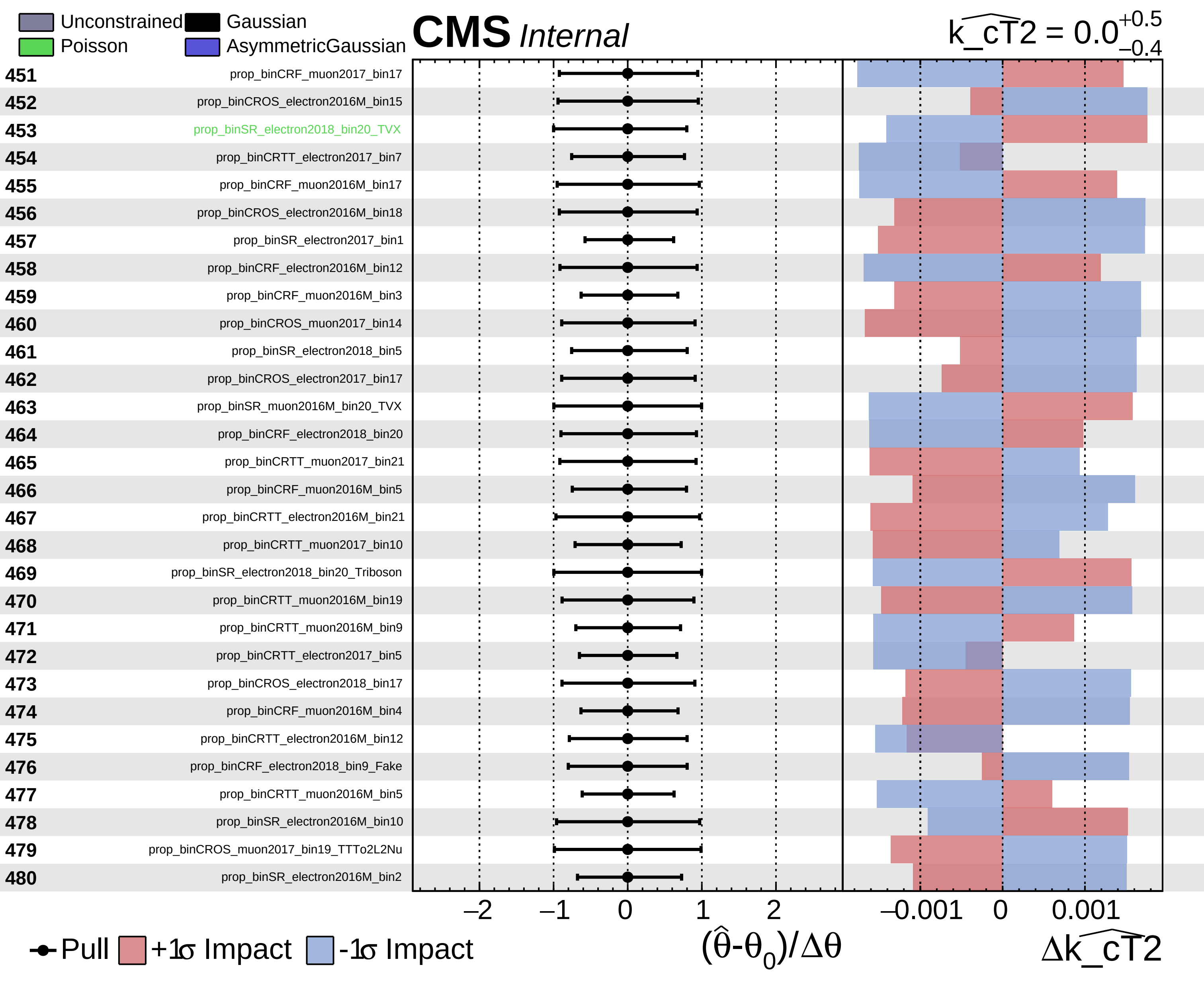

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
361 prop_binSR_muon2017_bin8
362 prop_binCROS_muon2016M_bin14
363 prop_binCRF_electron2016M_bin19
364 prop_binCROS_muon2018_bin19
365 prop_binCROS_muon2017_bin10
366 prop_binCRF_electron2017_bin19
367 prop_binCRTT_muon2016M_bin3
368 prop_binCRTT_muon2016M_bin20
369 prop_binCRF_muon2017_bin9
370 prop_binCRTT_electron2017_bin18
371 prop_binCROS_electron2016M_bin9
372 prop_binSR_electron2016M_bin1_Fake
373 prop_binCROS_muon2016M_bin6
374 prop_binCRF_electron2017_bin3
375 prop_binCROS_electron2018_bin8
376 prop_binCRTT_muon2016M_bin10
377 prop_binCRF_muon2016M_bin18_TTTo2L2Nu
378 prop_binSR_electron2018_bin9
379 prop_binCROS_muon2016M_bin11
380 prop_binCROS_electron2017_bin18
381 prop_binCRF_muon2017_bin14
382 prop_binCRTT_electron2018_bin17
383 prop_binSR_muon2016M_bin10
384 prop_binCROS_muon2017_bin15
385 prop_binSR_electron2017_bin10
386 prop_binCRF_electron2016M_bin21_sm
387 prop_binSR_electron2017_bin7
388 prop_binSR_electron2017_bin5_Fake
389 prop_binSR_electron2017_bin0
390 prop_binCROS_muon2016M_bin12
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



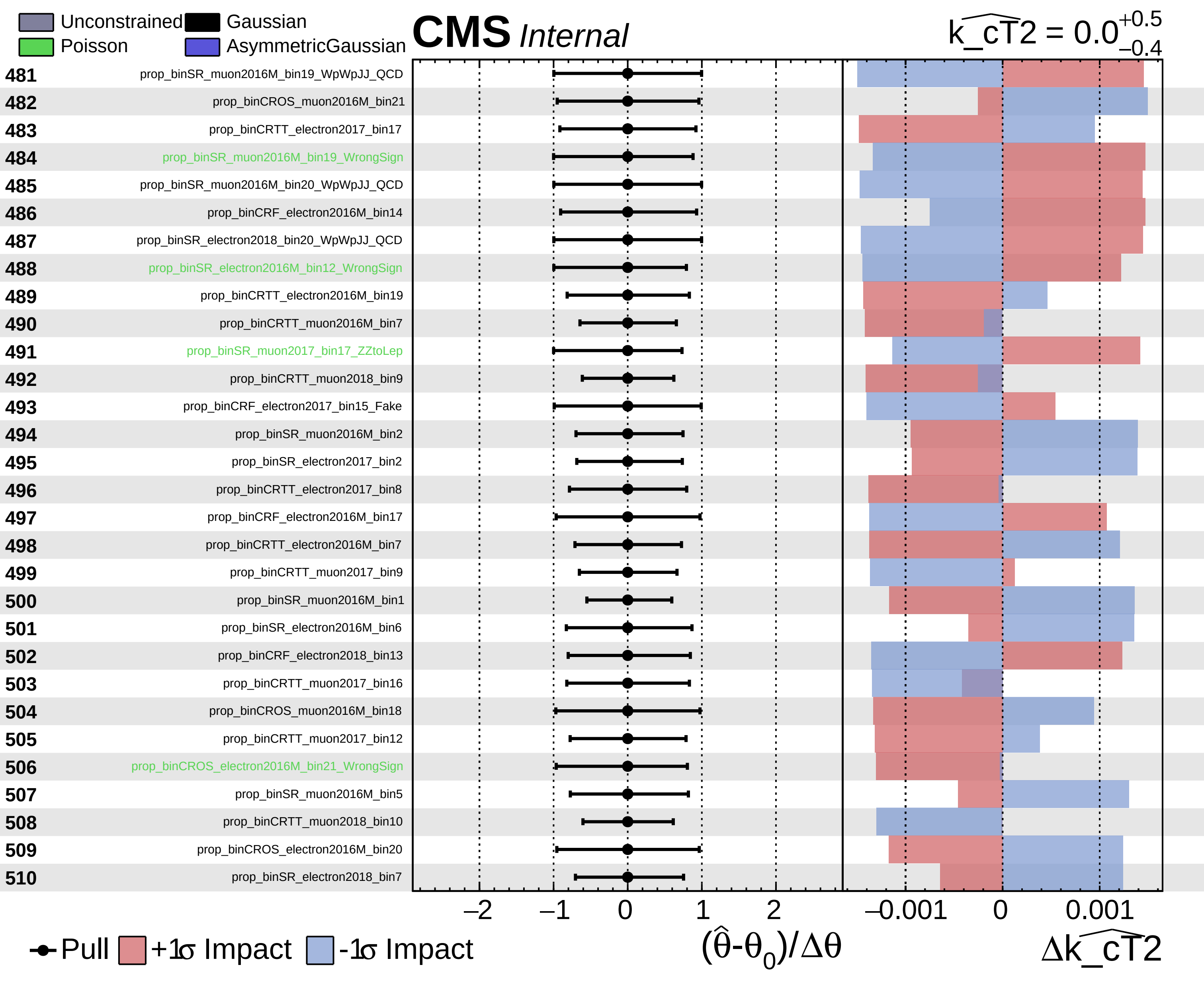
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
391 prop_binSR_muon2018_bin4
392 prop_binCROS_electron2018_bin5
393 prop_binCRTT_electron2016M_bin11
394 prop_binCRF_electron2018_bin5
395 prop_binCRTT_electron2017_bin12
396 prop_binCRTT_muon2016M_bin13
397 prop_binSR_electron2017_bin8
398 prop_binCRF_electron2016M_bin3
399 prop_binCROS_electron2018_bin20
400 prop_binCROS_muon2016M_bin19
401 prop_binCRTT_muon2018_bin12
402 prop_binSR_muon2017_bin9
403 prop_binCRTT_electron2018_bin16
404 prop_binSR_electron2017_bin5_WrongSign
405 prop_binCRTT_electron2018_bin11
406 prop_binCRF_muon2018_bin4
407 prop_binSR_electron2018_bin2
408 prop_binCRF_electron2016M_bin2
409 QCDScale_ZZtoLep
410 prop_binSR_electron2016M_bin11
411 prop_binCRF_electron2016M_bin6
412 prop_binCRTT_electron2016M_bin9
413 prop_binSR_electron2016M_bin4
414 prop_binCRTT_electron2018_bin7
415 prop_binCRF_electron2018_bin7
416 prop_binCROS_electron2016M_bin17_WrongSign
417 prop_binSR_muon2017_bin7
418 prop_binCROS_electron2016M_bin5
419 prop_binCRTT_electron2016M_bin17
420 prop_binCRF_electron2016M_bin13
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



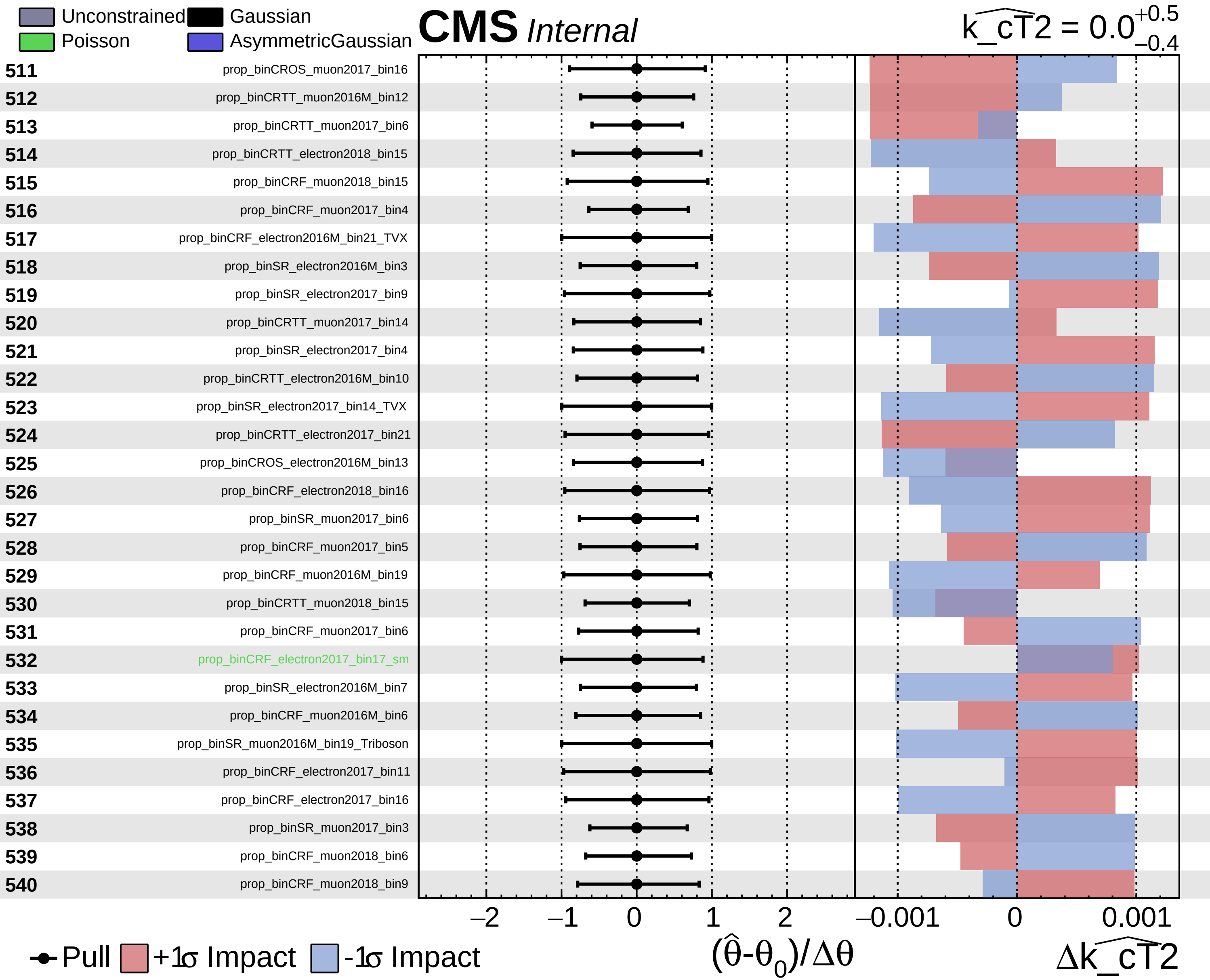
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
421 prop_binCRTT_electron2018_bin18
422 prop_binSR_muon2017_bin4
423 prop_binSR_electron2016M_bin20_TVX
424 prop_binCRTT_electron2016M_bin16
425 prop_binCROS_muon2017_bin11
426 prop_binSR_muon2016M_bin19_TVX
427 prop_binCRTT_muon2018_bin14
428 prop_binCRF_electron2017_bin20
429 prop_binCROS_electron2016M_bin3
430 prop_binCRF_muon2018_bin7
431 prop_binCRTT_muon2016M_bin6
432 prop_binCRTT_muon2016M_bin15
433 prop_binCRTT_muon2017_bin15
434 prop_binCRF_muon2018_bin18
435 prop_binCROS_electron2018_bin16
436 prop_binSR_muon2017_bin1
437 prop_binCRF_electron2018_bin8
438 prop_binSR_electron2018_bin10
439 prop_binCROS_muon2016M_bin20
440 prop_binCROS_electron2017_bin21
441 prop_binCRF_electron2017_bin5
442 prop_binCRF_muon2017_bin2
443 prop_binCRF_muon2018_bin12
444 prop_binSR_muon2017_bin2
445 prop_binCRF_muon2018_bin16
446 prop_binCRF_electron2018_bin6
447 prop_binCRTT_electron2016M_bin14
448 prop_binCRF_electron2016M_bin4
449 prop_binCRTT_muon2017_bin5
450 prop_binCRTT_electron2018_bin8
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



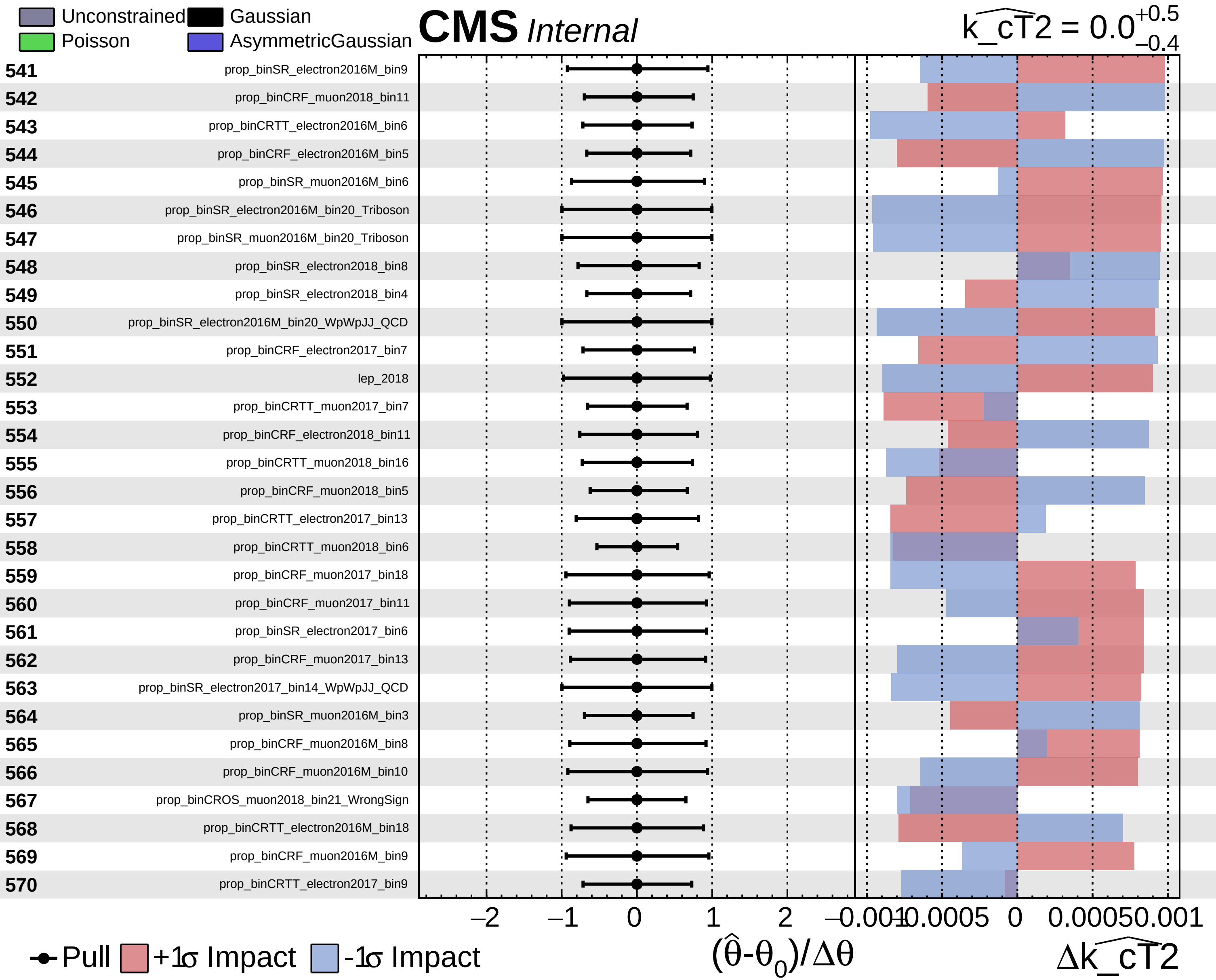
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
451 prop_binCRF_muon2017_bin17
452 prop_binCROS_electron2016M_bin15
453 prop_binSR_electron2018_bin20_TVX
454 prop_binCRTT_electron2017_bin7
455 prop_binCRF_muon2016M_bin17
456 prop_binCROS_electron2016M_bin18
457 prop_binSR_electron2017_bin1
458 prop_binCRF_electron2016M_bin12
459 prop_binCRF_muon2016M_bin3
460 prop_binCROS_muon2017_bin14
461 prop_binSR_electron2018_bin5
462 prop_binCROS_electron2017_bin17
463 prop_binSR_muon2016M_bin20_TVX
464 prop_binCRF_electron2018_bin20
465 prop_binCRTT_muon2017_bin21
466 prop_binCRF_muon2016M_bin5
467 prop_binCRTT_electron2016M_bin21
468 prop_binCRTT_muon2017_bin10
469 prop_binSR_electron2018_bin20_Triboson
470 prop_binCRTT_muon2016M_bin19
471 prop_binCRTT_muon2016M_bin9
472 prop_binCRTT_electron2017_bin5
473 prop_binCROS_electron2018_bin17
474 prop_binCRF_muon2016M_bin4
475 prop_binCRTT_electron2016M_bin12
476 prop_binCRF_electron2018_bin9_Fake
477 prop_binCRTT_muon2016M_bin5
478 prop_binSR_electron2016M_bin10
479 prop_binCROS_muon2017_bin19_TTTo2L2Nu
480 prop_binSR_electron2016M_bin2
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

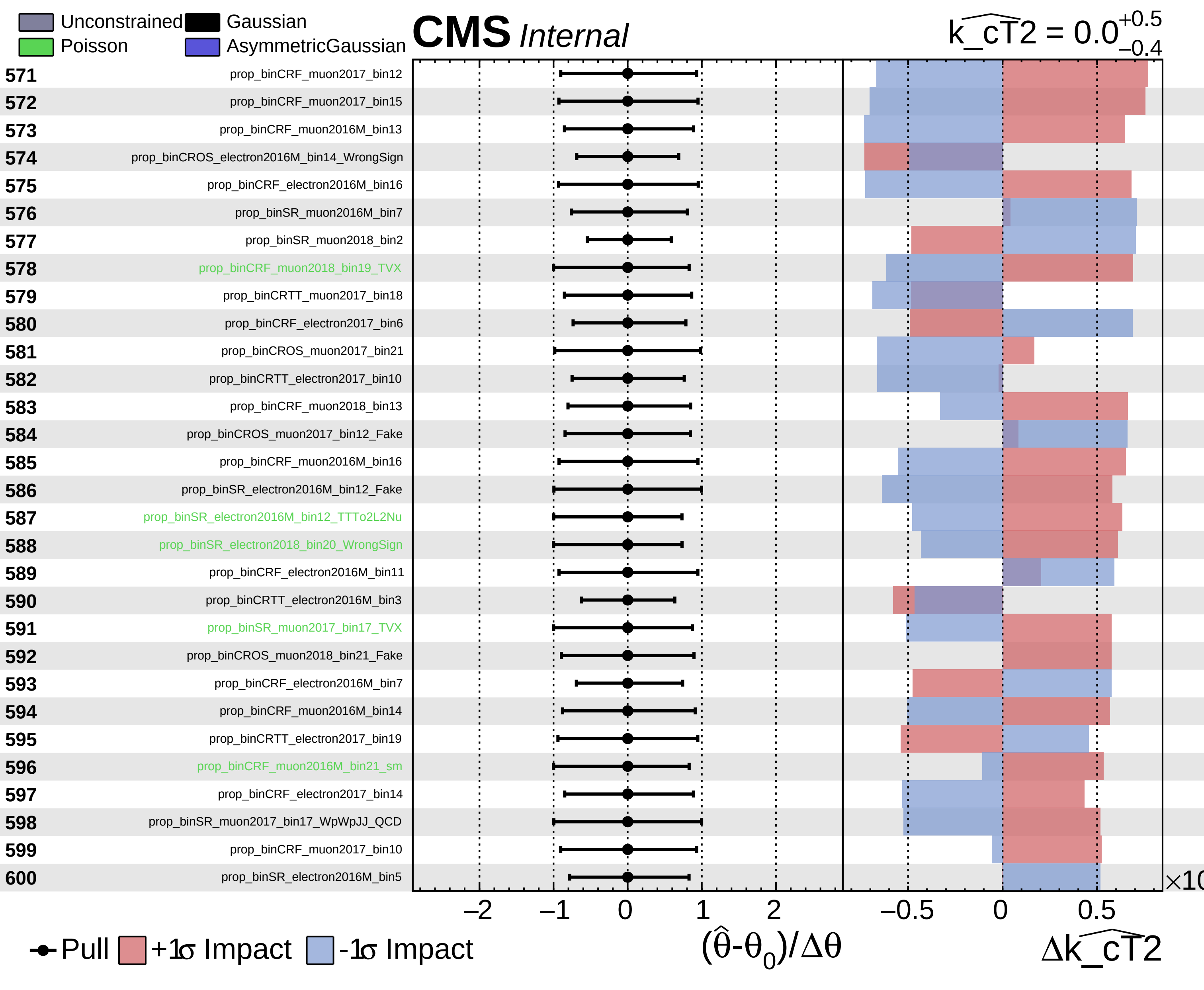


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
481 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
482 prop_binCROS_muon2016M_bin21
483 prop_binCRTT_electron2017_bin17
484 prop_binSR_muon2016M_bin19_WrongSign
485 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
486 prop_binCRF_electron2016M_bin14
487 prop_binSR_electron2018_bin20_WpWpJJ_QCD
488 prop_binSR_electron2016M_bin12_WrongSign
489 prop_binCRTT_electron2016M_bin19
490 prop_binCRTT_muon2016M_bin7
491 prop_binSR_muon2017_bin17_ZZtoLep
492 prop_binCRTT_muon2018_bin9
493 prop_binCRF_electron2017_bin15_Fake
494 prop_binSR_muon2016M_bin2
495 prop_binSR_electron2017_bin2
496 prop_binCRTT_electron2017_bin8
497 prop_binCRF_electron2016M_bin17
498 prop_binCRTT_electron2016M_bin7
499 prop_binCRTT_muon2017_bin9
500 prop_binSR_muon2016M_bin1
501 prop_binSR_electron2016M_bin6
502 prop_binCRF_electron2018_bin13
503 prop_binCRTT_muon2017_bin16
504 prop_binCROS_muon2016M_bin18
505 prop_binCRTT_muon2017_bin12
506 prop_binCROS_electron2016M_bin21_WrongSign
507 prop_binSR_muon2016M_bin5
508 prop_binCRTT_muon2018_bin10
509 prop_binCROS_electron2016M_bin20
510 prop_binSR_electron2018_bin7
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

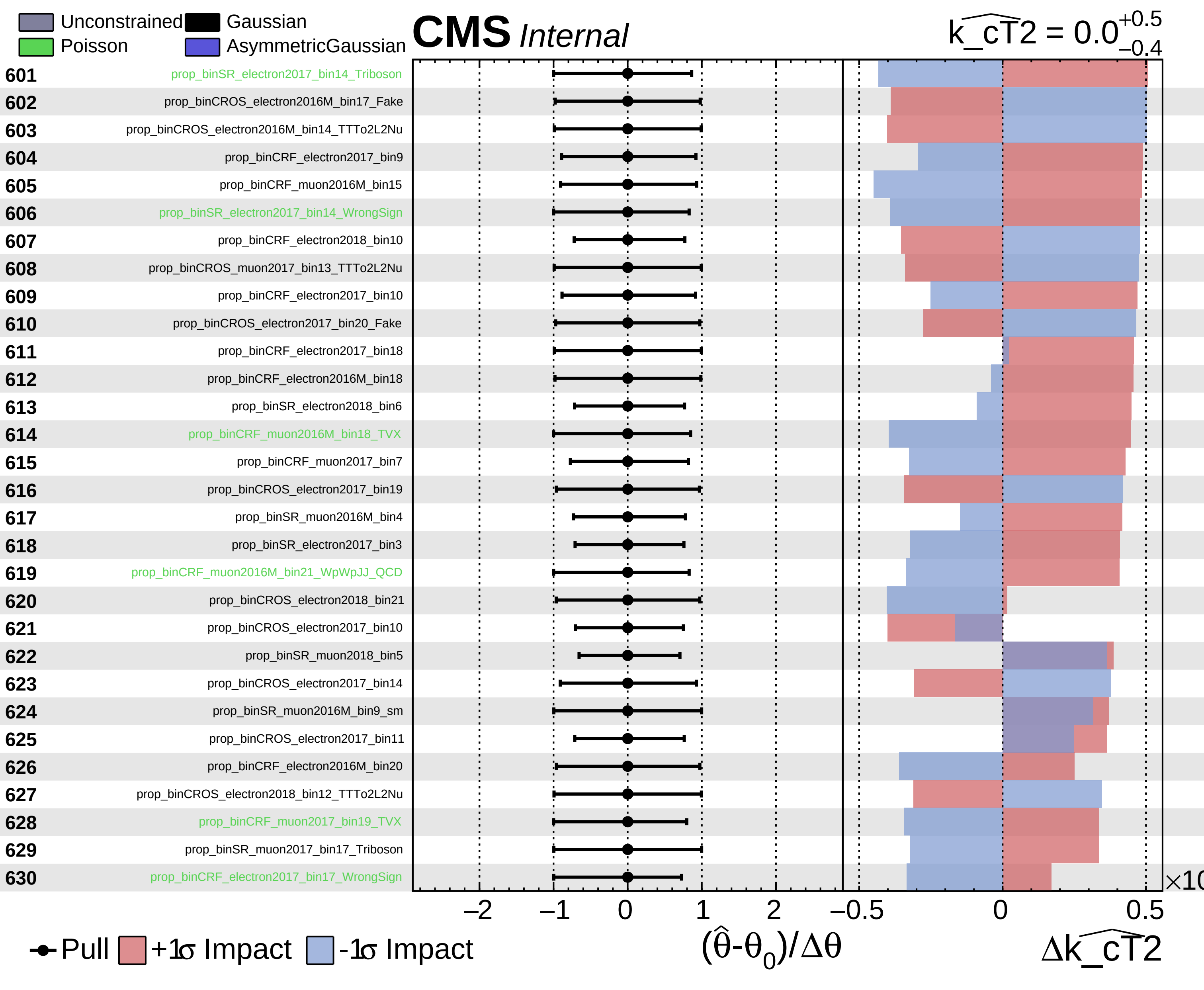


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
511 prop_binCROS_muon2017_bin16
512 prop_binCRTT_muon2016M_bin12
513 prop_binCRTT_muon2017_bin6
514 prop_binCRTT_electron2018_bin15
515 prop_binCRF_muon2018_bin15
516 prop_binCRF_muon2017_bin4
517 prop_binCRF_electron2016M_bin21_TVX
518 prop_binSR_electron2016M_bin3
519 prop_binSR_electron2017_bin9
520 prop_binCRTT_muon2017_bin14
521 prop_binSR_electron2017_bin4
522 prop_binCRTT_electron2016M_bin10
523 prop_binSR_electron2017_bin14_TVX
524 prop_binCRTT_electron2017_bin21
525 prop_binCROS_electron2016M_bin13
526 prop_binCRF_electron2018_bin16
527 prop_binSR_muon2017_bin6
528 prop_binCRF_muon2017_bin5
529 prop_binCRF_muon2016M_bin19
530 prop_binCRTT_muon2018_bin15
531 prop_binCRF_muon2017_bin6
532 prop_binCRF_electron2017_bin17_sm
533 prop_binSR_electron2016M_bin7
534 prop_binCRF_muon2016M_bin6
535 prop_binSR_muon2016M_bin19_Triboson
536 prop_binCRF_electron2017_bin11
537 prop_binCRF_electron2017_bin16
538 prop_binSR_muon2017_bin3
539 prop_binCRF_muon2018_bin6
540 prop_binCRF_muon2018_bin9
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

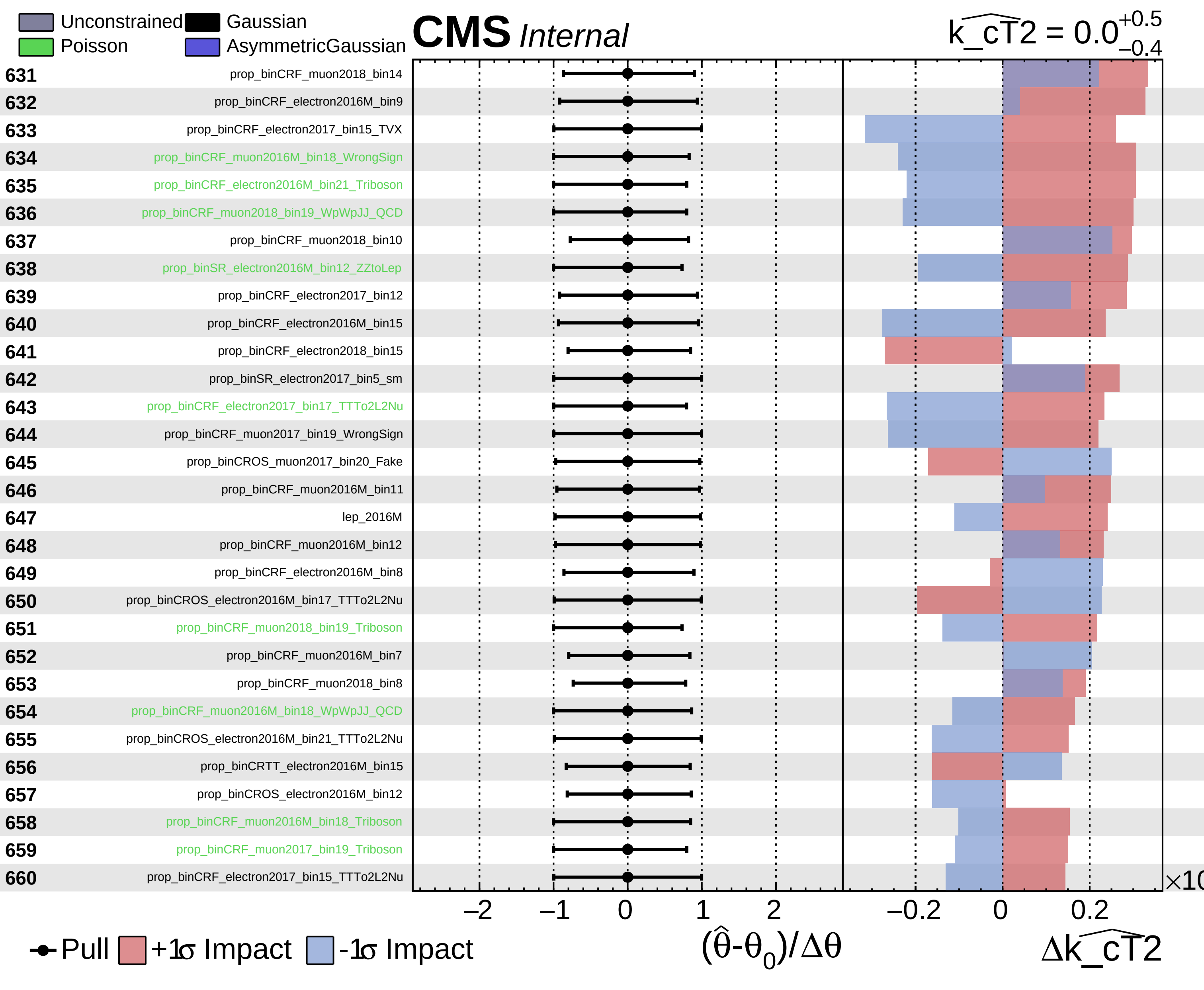

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
541 prop_binSR_electron2016M_bin9
542 prop_binCRF_muon2018_bin11
543 prop_binCRTT_electron2016M_bin6
544 prop_binCRF_electron2016M_bin5
545 prop_binSR_muon2016M_bin6
546 prop_binSR_electron2016M_bin20_Triboson
547 prop_binSR_muon2016M_bin20_Triboson
548 prop_binSR_electron2018_bin8
549 prop_binSR_electron2018_bin4
550 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
551 prop_binCRF_electron2017_bin7
552 lep_2018
553 prop_binCRTT_muon2017_bin7
554 prop_binCRF_electron2018_bin11
555 prop_binCRTT_muon2018_bin16
556 prop_binCRF_muon2018_bin5
557 prop_binCRTT_electron2017_bin13
558 prop_binCRTT_muon2018_bin6
559 prop_binCRF_muon2017_bin18
560 prop_binCRF_muon2017_bin11
561 prop_binSR_electron2017_bin6
562 prop_binCRF_muon2017_bin13
563 prop_binSR_electron2017_bin14_WpWpJJ_QCD
564 prop_binSR_muon2016M_bin3
565 prop_binCRF_muon2016M_bin8
566 prop_binCRF_muon2016M_bin10
567 prop_binCROS_muon2018_bin21_WrongSign
568 prop_binCRTT_electron2016M_bin18
569 prop_binCRF_muon2016M_bin9
570 prop_binCRTT_electron2017_bin9
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



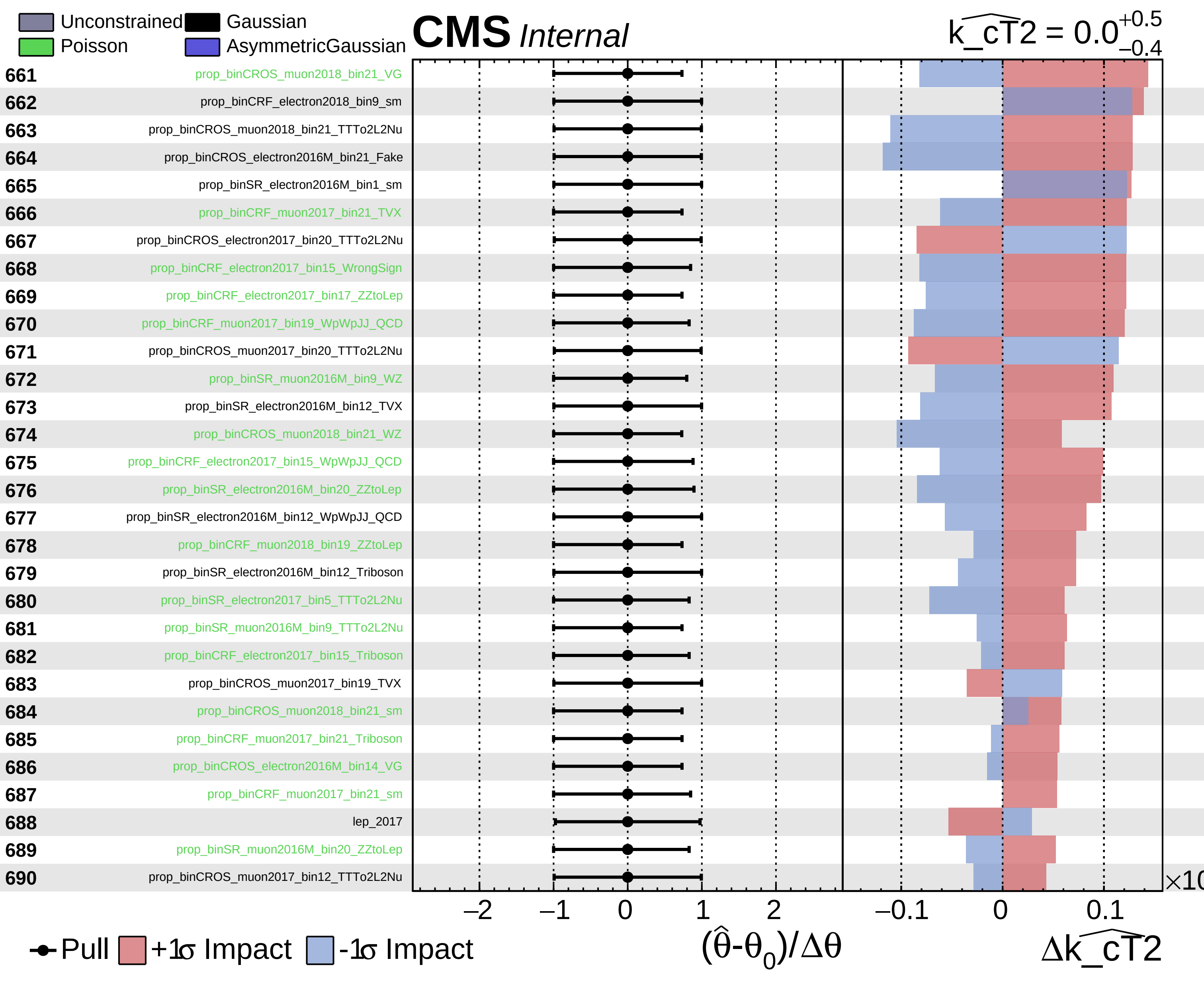
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
571 prop_binCRF_muon2017_bin12
572 prop_binCRF_muon2017_bin15
573 prop_binCRF_muon2016M_bin13
574 prop_binCROS_electron2016M_bin14_WrongSign
575 prop_binCRF_electron2016M_bin16
576 prop_binSR_muon2016M_bin7
577 prop_binSR_muon2018_bin2
578 prop_binCRF_muon2018_bin19_TVX
579 prop_binCRTT_muon2017_bin18
580 prop_binCRF_electron2017_bin6
581 prop_binCROS_muon2017_bin21
582 prop_binCRTT_electron2017_bin10
583 prop_binCRF_muon2018_bin13
584 prop_binCROS_muon2017_bin12_Fake
585 prop_binCRF_muon2016M_bin16
586 prop_binSR_electron2016M_bin12_Fake
587 prop_binSR_electron2016M_bin12_TTTo2L2Nu
588 prop_binSR_electron2018_bin20_WrongSign
589 prop_binCRF_electron2016M_bin11
590 prop_binCRTT_electron2016M_bin3
591 prop_binSR_muon2017_bin17_TVX
592 prop_binCROS_muon2018_bin21_Fake
593 prop_binCRF_electron2016M_bin7
594 prop_binCRF_muon2016M_bin14
595 prop_binCRTT_electron2017_bin19
596 prop_binCRF_muon2016M_bin21_sm
597 prop_binCRF_electron2017_bin14
598 prop_binSR_muon2017_bin17_WpWpJJ_QCD
599 prop_binCRF_muon2017_bin10
600 prop_binSR_electron2016M_bin5
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

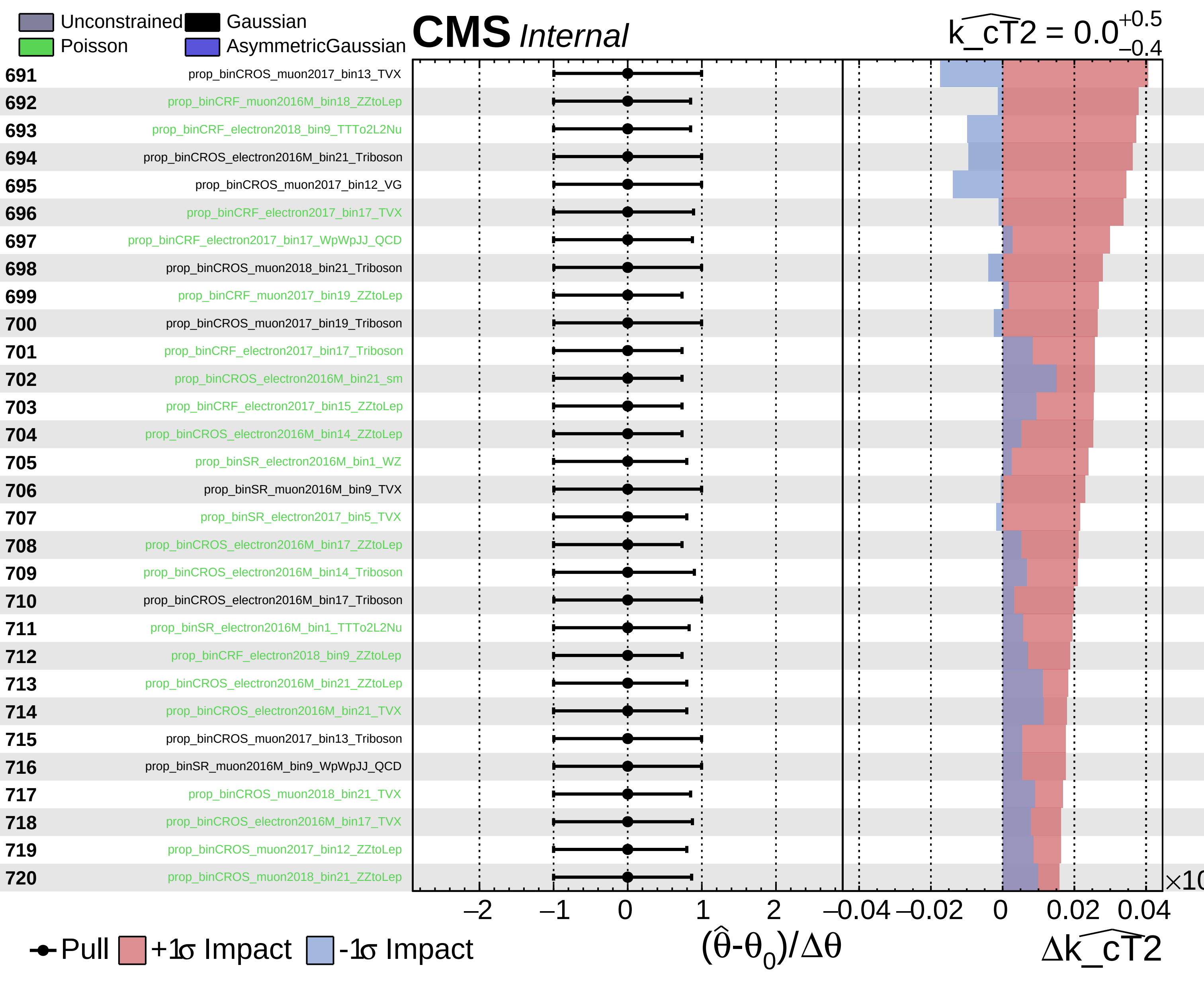


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
601 prop_binSR_electron2017_bin14_Triboson
602 prop_binCROS_electron2016M_bin17_Fake
603 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
604 prop_binCRF_electron2017_bin9
605 prop_binCRF_muon2016M_bin15
606 prop_binSR_electron2017_bin14_WrongSign
607 prop_binCRF_electron2018_bin10
608 prop_binCROS_muon2017_bin13_TTTo2L2Nu
609 prop_binCRF_electron2017_bin10
610 prop_binCROS_electron2017_bin20_Fake
611 prop_binCRF_electron2017_bin18
612 prop_binCRF_electron2016M_bin18
613 prop_binSR_electron2018_bin6
614 prop_binCRF_muon2016M_bin18_TVX
615 prop_binCRF_muon2017_bin7
616 prop_binCROS_electron2017_bin19
617 prop_binSR_muon2016M_bin4
618 prop_binSR_electron2017_bin3
619 prop_binCRF_muon2016M_bin21_WpWpJJ_QCD
620 prop_binCROS_electron2018_bin21
621 prop_binCROS_electron2017_bin10
622 prop_binSR_muon2018_bin5
623 prop_binCROS_electron2017_bin14
624 prop_binSR_muon2016M_bin9_sm
625 prop_binCROS_electron2017_bin11
626 prop_binCRF_electron2016M_bin20
627 prop_binCROS_electron2018_bin12_TTTo2L2Nu
628 prop_binCRF_muon2017_bin19_TVX
629 prop_binSR_muon2017_bin17_Triboson
630 prop_binCRF_electron2017_bin17_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

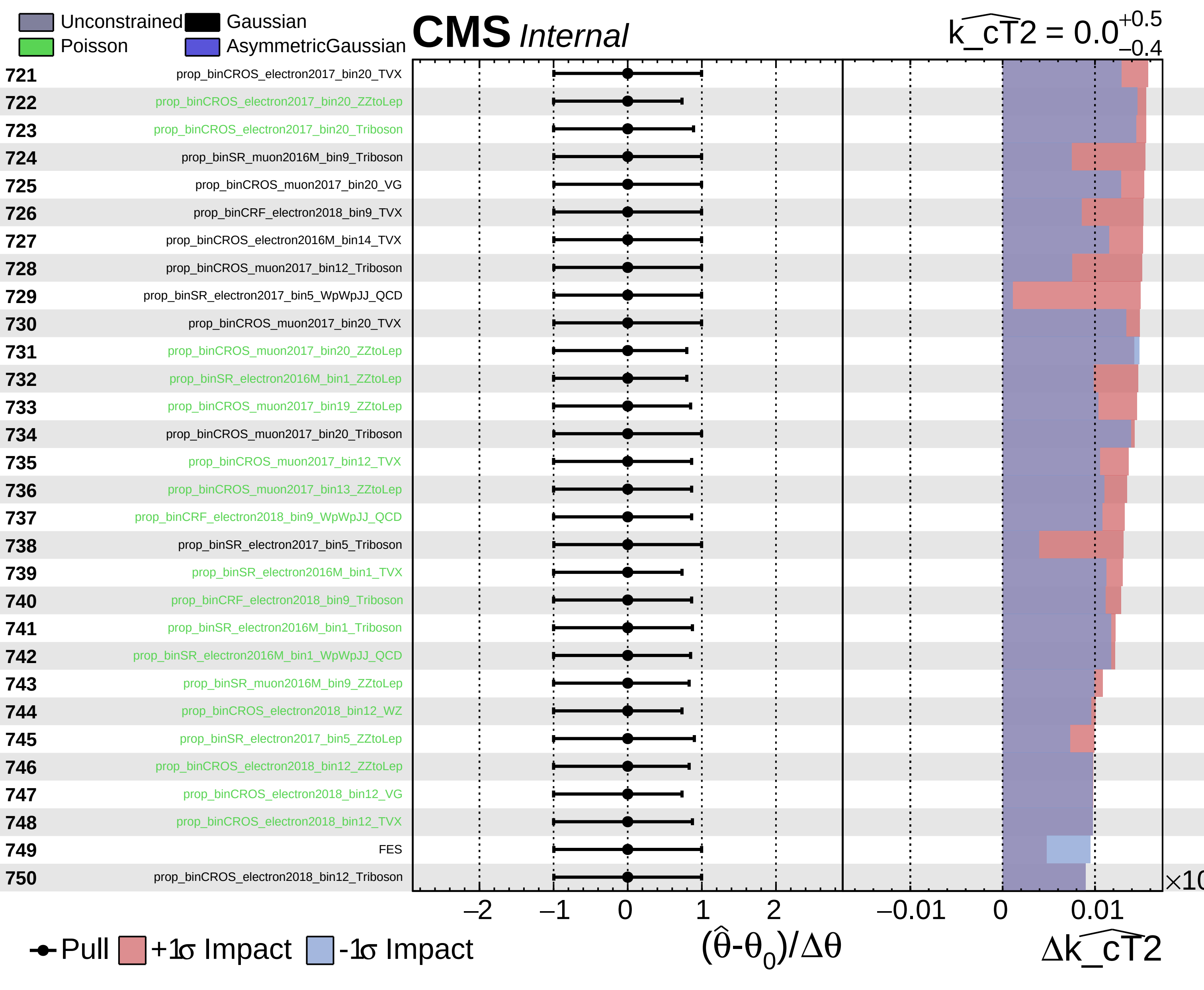


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
631 prop_binCRF_muon2018_bin14
632 prop_binCRF_electron2016M_bin9
633 prop_binCRF_electron2017_bin15_TVX
634 prop_binCRF_muon2016M_bin18_WrongSign
635 prop_binCRF_electron2016M_bin21_Triboson
636 prop_binCRF_muon2018_bin19_WpWpJJ_QCD
637 prop_binCRF_muon2018_bin10
638 prop_binSR_electron2016M_bin12_ZZtoLep
639 prop_binCRF_electron2017_bin12
640 prop_binCRF_electron2016M_bin15
641 prop_binCRF_electron2018_bin15
642 prop_binSR_electron2017_bin5_sm
643 prop_binCRF_electron2017_bin17_TTTo2L2Nu
644 prop_binCRF_muon2017_bin19_WrongSign
645 prop_binCROS_muon2017_bin20_Fake
646 prop_binCRF_muon2016M_bin11
647 lep_2016M
648 prop_binCRF_muon2016M_bin12
649 prop_binCRF_electron2016M_bin8
650 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
651 prop_binCRF_muon2018_bin19_Triboson
652 prop_binCRF_muon2016M_bin7
653 prop_binCRF_muon2018_bin8
654 prop_binCRF_muon2016M_bin18_WpWpJJ_QCD
655 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
656 prop_binCRTT_electron2016M_bin15
657 prop_binCROS_electron2016M_bin12
658 prop_binCRF_muon2016M_bin18_Triboson
659 prop_binCRF_muon2017_bin19_Triboson
660 prop_binCRF_electron2017_bin15_TTTo2L2Nu
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

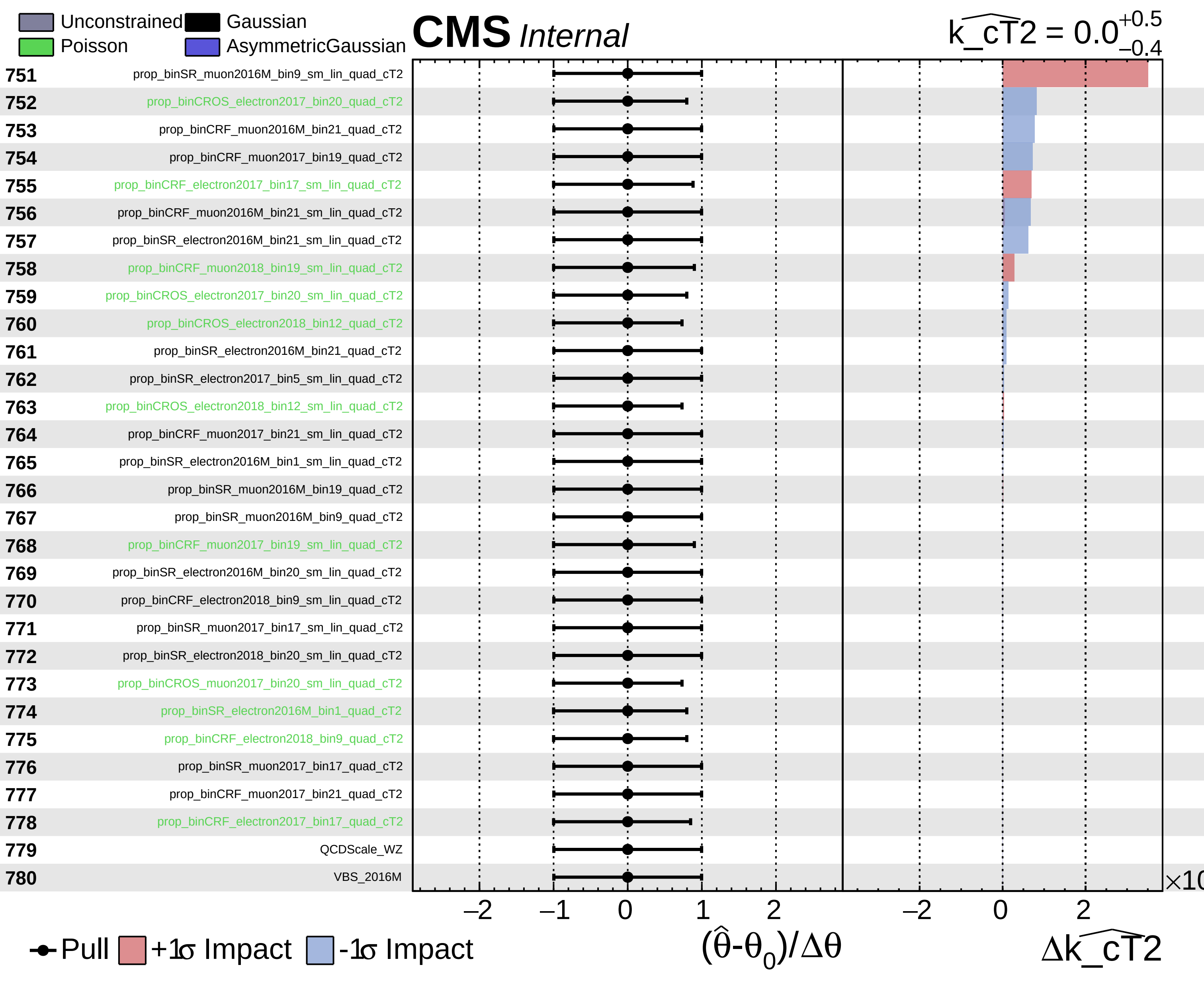

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
661 prop_binCROS_muon2018_bin21_VG
662 prop_binCRF_electron2018_bin9_sm
663 prop_binCROS_muon2018_bin21_TTTo2L2Nu
664 prop_binCROS_electron2016M_bin21_Fake
665 prop_binSR_electron2016M_bin1_sm
666 prop_binCRF_muon2017_bin21_TVX
667 prop_binCROS_electron2017_bin20_TTTo2L2Nu
668 prop_binCRF_electron2017_bin15_WrongSign
669 prop_binCRF_electron2017_bin17_ZZtoLep
670 prop_binCRF_muon2017_bin19_WpWpJJ_QCD
671 prop_binCROS_muon2017_bin20_TTTo2L2Nu
672 prop_binSR_muon2016M_bin9_WZ
673 prop_binSR_electron2016M_bin12_TVX
674 prop_binCROS_muon2018_bin21_WZ
675 prop_binCRF_electron2017_bin15_WpWpJJ_QCD
676 prop_binSR_electron2016M_bin20_ZZtoLep
677 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
678 prop_binCRF_muon2018_bin19_ZZtoLep
679 prop_binSR_electron2016M_bin12_Triboson
680 prop_binSR_electron2017_bin5_TTTo2L2Nu
681 prop_binSR_muon2016M_bin9_TTTo2L2Nu
682 prop_binCRF_electron2017_bin15_Triboson
683 prop_binCROS_muon2017_bin19_TVX
684 prop_binCROS_muon2018_bin21_sm
685 prop_binCRF_muon2017_bin21_Triboson
686 prop_binCROS_electron2016M_bin14_VG
687 prop_binCRF_muon2017_bin21_sm
688 lep_2017
689 prop_binSR_muon2016M_bin20_ZZtoLep
690 prop_binCROS_muon2017_bin12_TTTo2L2Nu
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



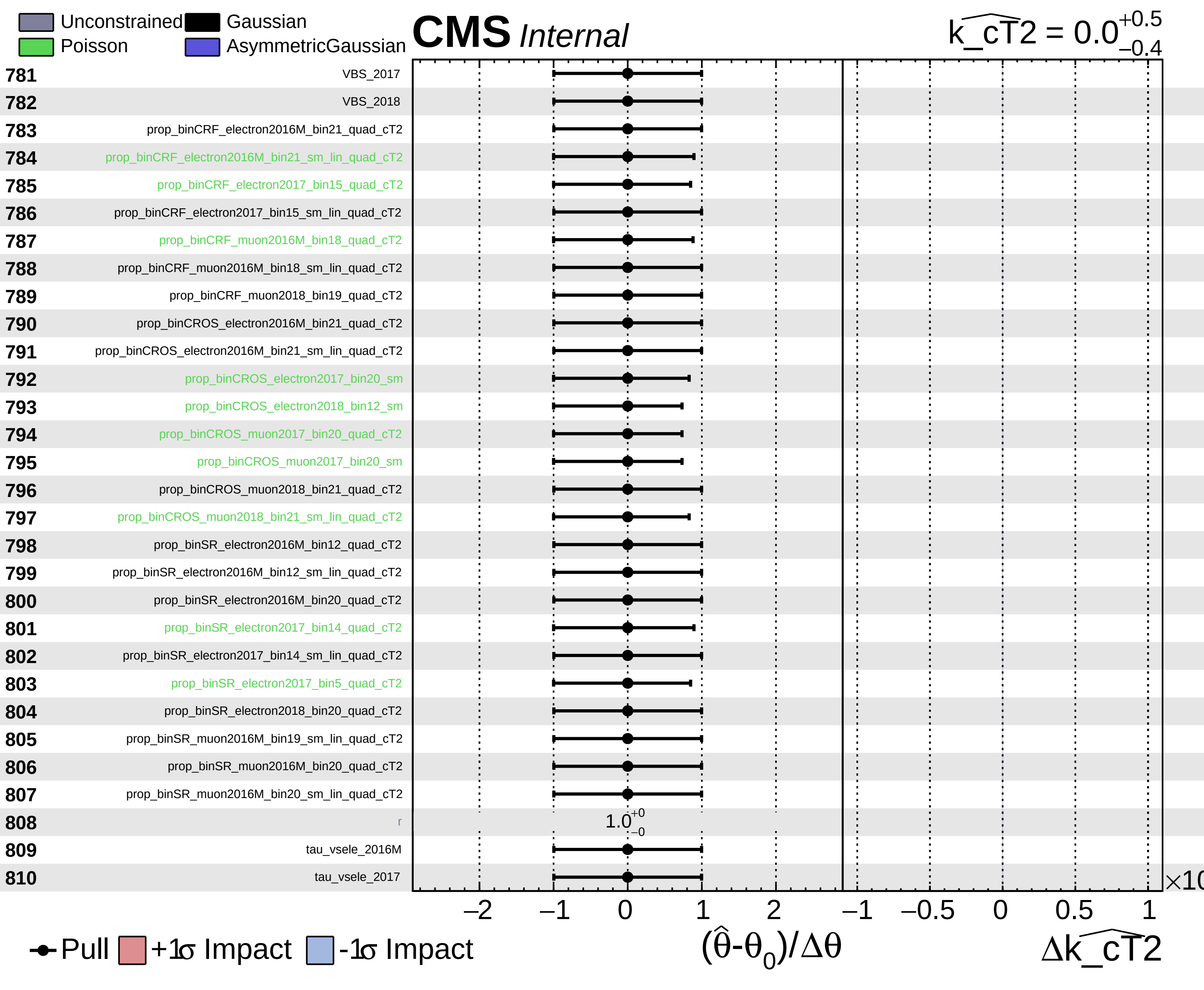
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
691 prop_binCROS_muon2017_bin13_TVX
692 prop_binCRF_muon2016M_bin18_ZZtoLep
693 prop_binCRF_electron2018_bin9_TTTo2L2Nu
694 prop_binCROS_electron2016M_bin21_Triboson
695 prop_binCROS_muon2017_bin12_VG
696 prop_binCRF_electron2017_bin17_TVX
697 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
698 prop_binCROS_muon2018_bin21_Triboson
699 prop_binCRF_muon2017_bin19_ZZtoLep
700 prop_binCROS_muon2017_bin19_Triboson
701 prop_binCRF_electron2017_bin17_Triboson
702 prop_binCROS_electron2016M_bin21_sm
703 prop_binCRF_electron2017_bin15_ZZtoLep
704 prop_binCROS_electron2016M_bin14_ZZtoLep
705 prop_binSR_electron2016M_bin1_WZ
706 prop_binSR_muon2016M_bin9_TVX
707 prop_binSR_electron2017_bin5_TVX
708 prop_binCROS_electron2016M_bin17_ZZtoLep
709 prop_binCROS_electron2016M_bin14_Triboson
710 prop_binCROS_electron2016M_bin17_Triboson
711 prop_binSR_electron2016M_bin1_TTTo2L2Nu
712 prop_binCRF_electron2018_bin9_ZZtoLep
713 prop_binCROS_electron2016M_bin21_ZZtoLep
714 prop_binCROS_electron2016M_bin21_TVX
715 prop_binCROS_muon2017_bin13_Triboson
716 prop_binSR_muon2016M_bin9_WpWpJJ_QCD
717 prop_binCROS_muon2018_bin21_TVX
718 prop_binCROS_electron2016M_bin17_TVX
719 prop_binCROS_muon2017_bin12_ZZtoLep
720 prop_binCROS_muon2018_bin21_ZZtoLep
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

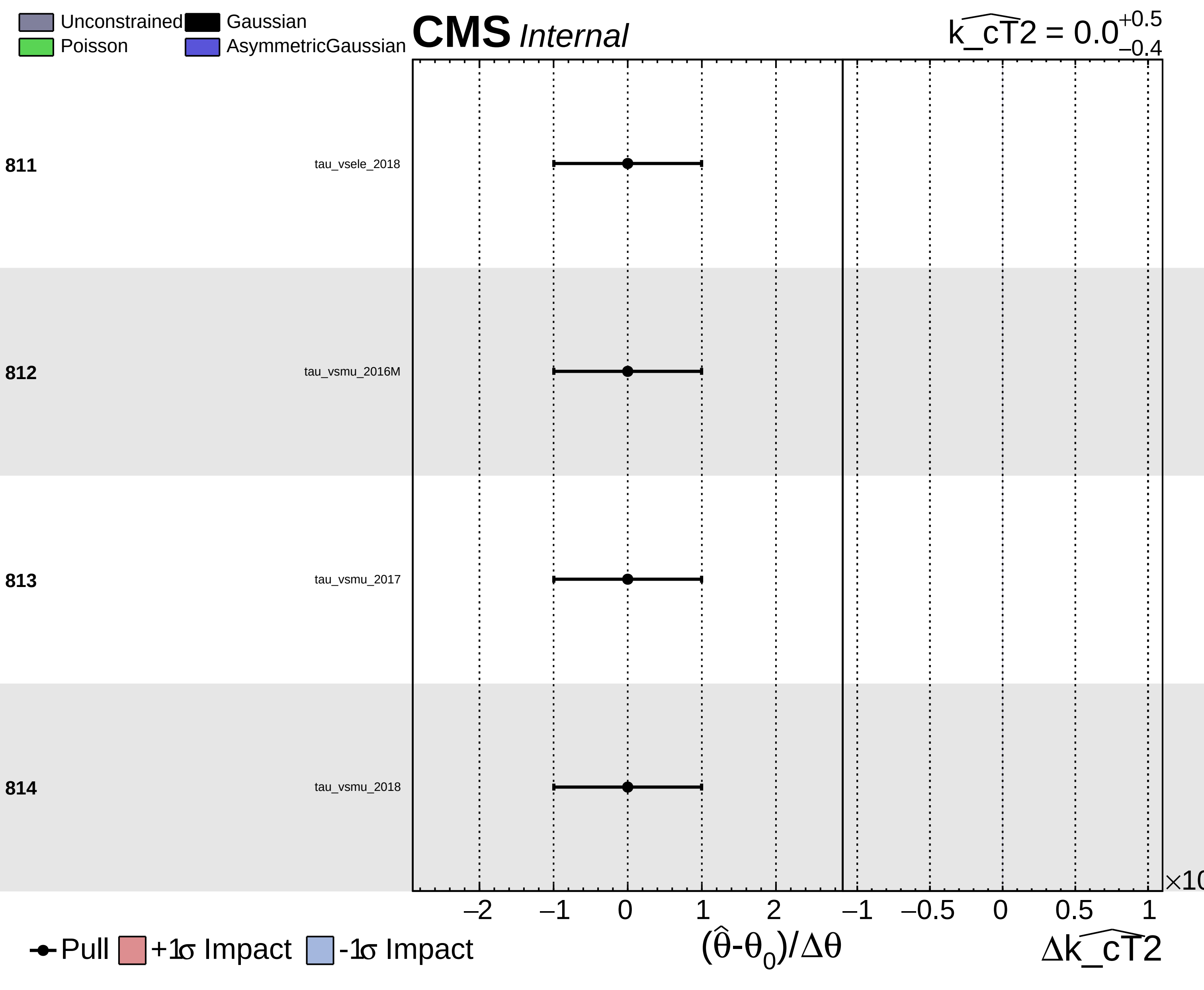


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
721 prop_binCROS_electron2017_bin20_TVX
722 prop_binCROS_electron2017_bin20_ZZtoLep
723 prop_binCROS_electron2017_bin20_Triboson
724 prop_binSR_muon2016M_bin9_Triboson
725 prop_binCROS_muon2017_bin20_VG
726 prop_binCRF_electron2018_bin9_TVX
727 prop_binCROS_electron2016M_bin14_TVX
728 prop_binCROS_muon2017_bin12_Triboson
729 prop_binSR_electron2017_bin5_WpWpJJ_QCD
730 prop_binCROS_muon2017_bin20_TVX
731 prop_binCROS_muon2017_bin20_ZZtoLep
732 prop_binSR_electron2016M_bin1_ZZtoLep
733 prop_binCROS_muon2017_bin19_ZZtoLep
734 prop_binCROS_muon2017_bin20_Triboson
735 prop_binCROS_muon2017_bin12_TVX
736 prop_binCROS_muon2017_bin13_ZZtoLep
737 prop_binCRF_electron2018_bin9_WpWpJJ_QCD
738 prop_binSR_electron2017_bin5_Triboson
739 prop_binSR_electron2016M_bin1_TVX
740 prop_binCRF_electron2018_bin9_Triboson
741 prop_binSR_electron2016M_bin1_Triboson
742 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
743 prop_binSR_muon2016M_bin9_ZZtoLep
744 prop_binCROS_electron2018_bin12_WZ
745 prop_binSR_electron2017_bin5_ZZtoLep
746 prop_binCROS_electron2018_bin12_ZZtoLep
747 prop_binCROS_electron2018_bin12_VG
748 prop_binCROS_electron2018_bin12_TVX
749 FES
750 prop_binCROS_electron2018_bin12_Triboson
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
751 prop_binSR_muon2016M_bin9_sm_lin_quad_cT2
752 prop_binCROS_electron2017_bin20_quad_cT2
753 prop_binCRF_muon2016M_bin21_quad_cT2
754 prop_binCRF_muon2017_bin19_quad_cT2
755 prop_binCRF_electron2017_bin17_sm_lin_quad_cT2
756 prop_binCRF_muon2016M_bin21_sm_lin_quad_cT2
757 prop_binSR_electron2016M_bin21_sm_lin_quad_cT2
758 prop_binCRF_muon2018_bin19_sm_lin_quad_cT2
759 prop_binCROS_electron2017_bin20_sm_lin_quad_cT2
760 prop_binCROS_electron2018_bin12_quad_cT2
761 prop_binSR_electron2016M_bin21_quad_cT2
762 prop_binSR_electron2017_bin5_sm_lin_quad_cT2
763 prop_binCROS_electron2018_bin12_sm_lin_quad_cT2
764 prop_binCRF_muon2017_bin21_sm_lin_quad_cT2
765 prop_binSR_electron2016M_bin1_sm_lin_quad_cT2
766 prop_binSR_muon2016M_bin19_quad_cT2
767 prop_binSR_muon2016M_bin9_quad_cT2
768 prop_binCRF_muon2017_bin19_sm_lin_quad_cT2
769 prop_binSR_electron2016M_bin20_sm_lin_quad_cT2
770 prop_binCRF_electron2018_bin9_sm_lin_quad_cT2
771 prop_binSR_muon2017_bin17_sm_lin_quad_cT2
772 prop_binSR_electron2018_bin20_sm_lin_quad_cT2
773 prop_binCROS_muon2017_bin20_sm_lin_quad_cT2
774 prop_binSR_electron2016M_bin1_quad_cT2
775 prop_binCRF_electron2018_bin9_quad_cT2
776 prop_binSR_muon2017_bin17_quad_cT2
777 prop_binCRF_muon2017_bin21_quad_cT2
778 prop_binCRF_electron2017_bin17_quad_cT2
779 QCDScale_WZ
780 VBS_2016M
−2
−1
0
1
2
−2
0
2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+0.5}_{-0.4}$
781 VBS_2017
782 VBS_2018
783 prop_binCRF_electron2016M_bin21_quad_cT2
784 prop_binCRF_electron2016M_bin21_sm_lin_quad_cT2
785 prop_binCRF_electron2017_bin15_quad_cT2
786 prop_binCRF_electron2017_bin15_sm_lin_quad_cT2
787 prop_binCRF_muon2016M_bin18_quad_cT2
788 prop_binCRF_muon2016M_bin18_sm_lin_quad_cT2
789 prop_binCRF_muon2018_bin19_quad_cT2
790 prop_binCROS_electron2016M_bin21_quad_cT2
791 prop_binCROS_electron2016M_bin21_sm_lin_quad_cT2
792 prop_binCROS_electron2017_bin20_sm
793 prop_binCROS_electron2018_bin12_sm
794 prop_binCROS_muon2017_bin20_quad_cT2
795 prop_binCROS_muon2017_bin20_sm
796 prop_binCROS_muon2018_bin21_quad_cT2
797 prop_binCROS_muon2018_bin21_sm_lin_quad_cT2
798 prop_binSR_electron2016M_bin12_quad_cT2
799 prop_binSR_electron2016M_bin12_sm_lin_quad_cT2
800 prop_binSR_electron2016M_bin20_quad_cT2
801 prop_binSR_electron2017_bin14_quad_cT2
802 prop_binSR_electron2017_bin14_sm_lin_quad_cT2
803 prop_binSR_electron2017_bin5_quad_cT2
804 prop_binSR_electron2018_bin20_quad_cT2
805 prop_binSR_muon2016M_bin19_sm_lin_quad_cT2
806 prop_binSR_muon2016M_bin20_quad_cT2
807 prop_binSR_muon2016M_bin20_sm_lin_quad_cT2
808 r
1.0^{+0}_{-0}
809 tau_vsele_2016M
810 tau_vsele_2017
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +0.5 −0.4
811 tau_vsele_2018
812 tau_vsmu_2016M
813 tau_vsmu_2017
814 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2